

BSCS FINAL PROJECT

Software Design Specification

ReMIND: Generative AI Based Memory Rebuilder



Zulkifl

Project Advisor

Muhammad Zulkifl Hasan

Presented by:

Group ID: F25CS104

Registration No:

L1F22BSCS0086

L1F22BSCS0103

L1F22BSCS0094

Name:

M. Hassan Ashraf

M. Talha Faizan

Mubashar Tanveer

Faculty of Information Technology & Computer Science

University of Central Punjab

Software Design Specification

SDP Phase II

ReMIND: Generative AI based Memory Rebuilder

Advisor: Muhammad Zulkifl Hasan

Team F25CS0104

Member Name	Primary Responsibility
M Hassan Ashraf	Project Management & Planning
M Talha Faizan	Figma & Designs
Mubashar Tanveer	Documentation & Review

Table of Contents

Revision History	ii
Previous Phases Feedback.....	ii
Abstract.....	iii
1. Introduction.....	1
1.1 Product.....	1
1.2 Background.....	1
1.3 Objective(s)/Aim(s)/Target(s)	1
1.4 Scope	2
1.5 Business Goals.....	2
1.6 Document Conventions	2
2. Overall Description	3
2.1 Product Features	3
2.2 Functional Description	3
2.3 User Classes and Characteristics	5
2.4 Design and Implementation Constraints.....	5
2.5 Assumptions and Dependencies	7
3. Technical Architecture	8
3.1 Application and Data Architecture	10
3.2 Component Interactions and Collaborations	14
3.3 Design Reuse and Design Patterns	21
3.4 Technology Architecture	21
Data and Processing Characteristics	21
Technology Stack and Platforms	22
4. Screenshots/Prototype	23
4.1 Workflow.....	23
4.2 Screens.....	24
4.3 Additional Information	32
5. Revised Project Plan	32
6. References	33
Appendix A: Glossary.....	35
Appendix B: IV & V Report	35

Revision History

Name	Date	Reason For Changes	Version
Idea Defense	September 11, 2025	Problem statements and scope are not clear.	Version 2

Previous Phases Feedback

Idea Defence Feedback (Screenshot)

F25 Idea Defence Feedback



Project Office

☺

↩ Reply

↩ Reply all

➡ Forward

📧

📎

📅

⋮

To: 📧 L1F22BSCS0086- MUHAMMAD HASSAN ASHRAF; 📧 L1F22BSCS0103- MUHAMMAD TALHA FAIZAN; +1 other Mon 01/09/2025 10:35

Cc: 📧 Dr Adnan Ghafoor; 🟢 Muhammad Zulkifli Hasan

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	CS-15
Project Title	GENERATIVE AI BASED MEMORY REBUILDER

Similar products already exist. If the patient/a family member also does not have the details of event which we want to restore. Then system cannot perform.

Problem statement not clear. Scope not clear.

Students are incapable of implementing the idea. They don't know the basics of the project. Better to change the idea.

Major revision required. Revise your project idea in light of comments provided. Last date to submit the revised idea form is **Thursday, September 11, 2025, before 3:00 PM** in Project Office (F-304 Building A).

Abstract

ReMIND is an AI assisted web application designed to support individuals experiencing memory loss disorders such as Alzheimer’s disease, dementia, and age-related cognitive decline. Memory loss affects not only a person’s ability to recall past events but also their emotional stability, relationships, and overall sense of identity. Most existing digital solutions rely heavily on manual input, and when no prior record or recollection exists, they fail to assist patients and caregivers in restoring meaningful experiences. ReMIND addresses this limitation by offering semi-automated memory reconstruction that helps users reconnect with their personal history.

The system collects and analyzes different forms of data, including written notes, photographs, and voice recordings, and organizes them into personalized timelines that reflect moments from daily life. Through artificial intelligence, ReMIND generates contextual details that can suggest likely places, people, and events related to the user’s memories. This enriched information helps patients retrieve forgotten experiences in a more natural and emotionally resonant way. The platform creates an interactive environment where patients and caregivers can revisit, discuss, and refine reconstructed memories together, strengthening emotional connection and supporting cognitive engagement.

The project brings together concepts from artificial intelligence, natural language processing, computer vision, speech processing, data management, and human centered design. The focus is not solely on automation but on creating an experience that prioritizes empathy, accessibility, and user comfort. By transforming fragmented or incomplete data into meaningful memory narratives, ReMIND aims to restore a sense of identity and continuity for individuals facing cognitive decline.

The expected outcome is a functional prototype that demonstrates how intelligently guided memory reconstruction can contribute to cognitive rehabilitation and long-term digital memory preservation. ReMIND highlights the potential of emerging AI systems to support emotional wellbeing and provide hope for individuals whose memories are fading.

1. Introduction

ReMIND is an AI-assisted web application aimed at helping individuals with memory loss, such as Alzheimer's and dementia patients, relive past experiences. It reconstructs fragmented memories using multimedia inputs. The project is motivated by the emotional and cognitive challenges faced by patients, emphasizing empathy, recall, and caregiver support through technology.

1.1 Product

ReMIND is an intelligent, web-based memory reconstruction system designed to assist individuals suffering from memory loss disorders such as Alzheimer's and dementia. The software addresses the limitations of existing reminder tools that rely heavily on manual event logging or user recall, which often fail when sufficient details are unavailable. ReMIND integrates multiple data sources such as photos, text notes, and voice recordings to automatically identify, organize, and recreate meaningful events. Using artificial intelligence and contextual analysis, the system reconstructs daily timelines and generates narrative memories that patients and caregivers can revisit. Its user-friendly, elderly-oriented interface ensures accessibility and comfort, while secure backend services manage data storage and processing. The system provides collaborative features for caregivers, enabling them to contribute and review reconstructed memories. Ultimately, ReMIND bridges emotional and cognitive gaps, improving quality of life by transforming fragmented inputs into coherent, emotionally engaging experiences that support recall and human connection.

1.2 Background

The project lies within the domains of artificial intelligence, cognitive rehabilitation, and healthcare technology. It focuses on applying AI techniques such as natural language processing, computer vision, and speech processing to assist individuals with memory loss. By merging technology and empathy, ReMIND enhances memory recall, emotional well-being, and caregiver collaboration.

1.3 Objective(s)/Aim(s)/Target(s)

The primary objective of ReMIND is to develop an AI-assisted web application that helps individuals with memory loss to reconstruct meaningful memories from fragmented multimedia inputs. The project aims to combine technology and empathy to create a system that supports both patients and caregivers in recalling and preserving life experiences.

The specific objectives are as follows:

Memory Reconstruction: Transform incomplete data such as text notes, photos, and voice recordings into coherent and contextually rich memory narratives.

Event Clustering: Use AI techniques to organize related media and information into structured timelines representing daily or significant events.

Reminder and Notification Module: Provide timely reminders and alerts to assist patients in maintaining routine and connection with daily life.

Elderly-Friendly Interface: Design a simple, accessible interface optimized for older users and non-technical caregivers.

Secure Data Handling: Implement encryption, authentication, and cloud storage to protect sensitive personal data.

Caregiver Collaboration: Enable caregivers to contribute and review reconstructed memories to ensure accuracy and emotional value.

Proof of Concept Demonstration: Deliver a functional prototype using existing AI APIs and web technologies to validate the concept.

Overall, ReMIND seeks to enhance emotional well-being, cognitive support, and human connection through accessible AI innovation.

1.4 Scope

The scope of ReMIND covers the design and development of a web-based application that reconstructs and presents memories using multimedia inputs such as text, images, and audio. The system includes event clustering, memory timeline generation, and a reminder module for patients and caregivers. It focuses on enhancing recall and emotional connection through AI-powered contextual analysis. The application will feature an elderly-friendly interface, secure data handling, and cloud-based deployment. Medical diagnosis, hospital EHR integration, and real-time biometric tracking are excluded from the project. It will be a Concept-of-Proof limited project.

1.5 Business Goals

As a proof-of-concept prototype, ReMIND aims to demonstrate the commercial potential of AI-driven memory reconstruction in healthcare and assisted living. Its primary goal is to validate the feasibility of using artificial intelligence for emotional and cognitive support in patients with memory loss. The project establishes a foundation for future development into a scalable product that could benefit healthcare institutions, caregivers, and families. By integrating accessible technology with empathetic design, ReMIND highlights opportunities for collaboration between AI developers and healthcare providers, showcasing how innovation can lead to meaningful, socially impactful digital health solutions in memory rehabilitation.

1.6 Document Conventions

The Document Conventions being used throughout the document are following:

- Text being used is Times New Roman.
- Main Headings are of Size 16.
- Sub-Headings are of Size 14.
- Content/Paragraph and Tertiary headings are of Size 12.
- Auto line spacing is used.

2. Overall Description

2.1 Product Features

ReMIND is designed as an AI-powered, web-based memory reconstruction system that integrates multimedia data text, images, and audio to help individuals with memory loss to relive and preserve meaningful life events. The product emphasizes accessibility, emotional relevance, and technological feasibility.

Key features include:

Memory Reconstruction Engine: Uses AI models to analyze multimedia inputs and generate coherent, context-rich narratives that rebuild fragmented memories.

Event Clustering and Timeline Generation: Automatically organizes related media and data into daily or event-based timelines, enabling users to view reconstructed memories chronologically.

Multimedia Enhancement: Employs tools such as Real-ESRGAN for image enhancement, Whisper for voice transcription, and ElevenLabs for realistic voice synthesis.

Interactive Chat Interface: Provides a user-friendly conversational interface where patients and caregivers can interact, view, and refine reconstructed memories.

Reminder and Notification Module: Assists patients with daily activities and event tracking through customizable reminders.

Caregiver Collaboration: Allows caregivers to contribute to additional context or verify memory accuracy.

Secure Cloud Storage: Ensures privacy and safety through encrypted data handling and cloud-based management.

Overall, ReMIND combines artificial intelligence, empathy, and intuitive design to transform scattered personal data into emotionally significant digital experiences that aid cognitive rehabilitation and emotional well-being.

2.2 Functional Description

The ReMIND platform provides an elderly-friendly digital experience built using React.js and CSS3, designed to support users with memory loss in recalling and organizing important events. The front-end enables users to input memories through text, speech, or images, set up daily reminders, and view reconstructed stories or restored images in a simple and accessible layout. All interface components are optimized for readability, high contrast visibility, and ease of navigation to ensure older adults can interact with the system comfortably.

On the backend, the system is powered by Flask or FastAPI, providing a secure and efficient communication layer between the user interface, the database, and the AI modules. The backend manages API requests, authentication, event clustering logic, and the overall orchestration of user data. It ensures that memory inputs, reminders, and media files are processed reliably while maintaining fast and responsive interactions with the frontend.

AI integration forms the core functionality of ReMIND. The system connects with external APIs for story generation, speech transcription, voice synthesis, and image restoration. These AI services transform raw user inputs into meaningful reconstructed memories, personalized narratives, and enhanced media. The integration layer also validates, processes, and formats the AI output before sending it back to the frontend for user viewing.

The platform uses PostgreSQL to store structured memory data such as dates, descriptions, and event metadata, while Firebase or Cloudinary handles media storage for images and audio files. This ensures both efficient data management and scalable media handling. A reminder and notification module is also incorporated, allowing users and caregivers to receive timely alerts based on memory schedules or important events.

ReMIND is deployed using platforms such as Render for the frontend and Heroku for the backend, ensuring accessibility and stable performance. A basic CI/CD structure automates updates and maintains system reliability. Finally, the system includes a comprehensive testing and documentation process, ensuring that all features from memory reconstruction to reminder delivery are validated for usability, accuracy, and accessibility before final deployment.

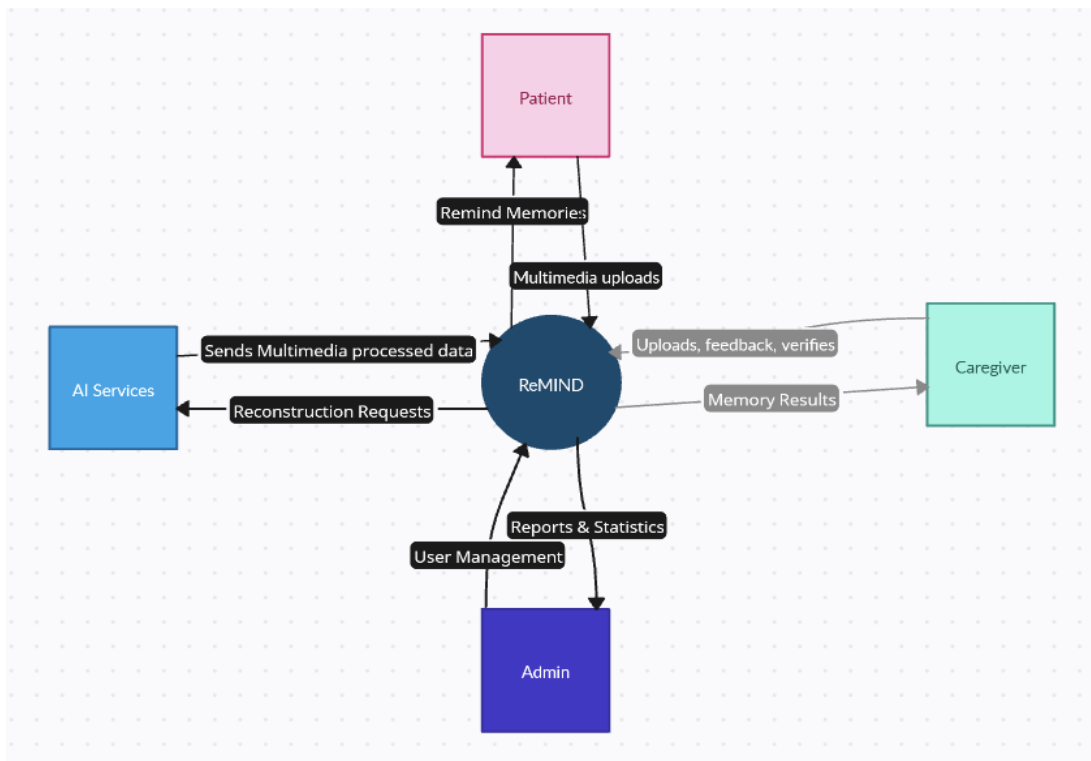


Fig 2.1: Context Diagram

2.3 User Classes and Characteristics

ReMIND is designed to serve two primary user classes **Patients (Memory Loss Individuals)** and **Caregivers (Family Members or Healthcare Assistants)** along with two secondary user classes System Administrators and Researchers/Developers. Each user class interacts with the system at different levels of complexity, privilege, and frequency.

1. **Patients (Primary Users):**
These are individuals affected by Alzheimer's, dementia, or general memory impairment. They use ReMIND frequently for viewing, listening to, and interacting with reconstructed memories or receiving reminders. The interface is designed to be elderly-friendly, visually clear, and minimal in complexity. Accessibility features such as large fonts, audio narration, and simplified navigation are prioritized.
2. **Caregivers (Primary Users):**
Family members, nurses, or attendants assist patients in managing and verifying reconstructed memories. They have moderate technical expertise and access privileges that allow them to upload multimedia inputs, annotate events, and review memory accuracy. Their primary goal is to monitor progress, reduce patient stress, and maintain emotional connections.
3. **System Administrators (Secondary Users):**
Administrators manage backend operations, including user account verification, data security, and cloud services. They ensure system reliability and compliance with ethical data standards.
4. **Researchers/Developers (Secondary Users):**
They may use ReMIND as a testbed to explore improvements in AI-based cognitive rehabilitation. Their access focuses on algorithmic evaluation and performance monitoring.

Favored user classes are **patients** and **caregivers**, as the system is primarily designed around their emotional and cognitive needs, while other classes ensure technical and operational continuity.

2.4 Design and Implementation Constraints

The evolution of the ReMIND platform is also subjected to a number of limitations that restrict the possible design options, as well as shaping the implementation options. The causes of these constraints are hardware availability, institutional policies, resource constraints, third-party latencies, and forced technology choices.

Hardware Limitations: The hardware is mainly based on the personal laptops of the team members, which are of a medium level, which do not have dedicated GPUs and thus are unable to perform intense AI tasks at the edge, e.g. restoring high-resolution images, neural text generation, or speech processing. Consequently, the AI modules are to be based on the cloud APIs or the high-performance GPU laboratories of UCP. The access to the UCP lab is limited by schedules, usage of slots, and availability of laboratories, which reduces time available to developers to execute heavy processing activities. Such constraints inhibit the scale of real-time AI processing and have to carefully distribute the work between local systems and cloud-based resources.

Software and Technology Limitations: The project requires the application of certain technologies as stipulated in the project scope. React.js should be used to create the frontend and Tailwind CSS, and Flask/FastAPI should be used to create the backend. Structured data will be stored using PostgreSQL, and Firebase/Cloudinary will be needed to utilize and store the media. Such pre-established technology stacks restrict the ability to use other tools (e.g. Django, MERN Stack, AWS S3). Likewise, the entire AI work should be based on external APIs, like Whisper, Real-ESRGAN, OpenAI, or Google Vision.

Such dependency limits the ability to customize and add complexity to integration and necessitates the need to conform to API-specific formats and rate limits.

Cloud Dependency & API Third Party: One of its biggest limitations is the use of external AI APIs to generate stories, speech-to-text, text-to-speech, and restore images. These APIs have use restrictions, request quotas, and latency limits and in some cases, paid subscriptions. The use of APIs such as Whisper or Eleven Labs can be costly to use on a regular basis. The system should therefore optimize the API calls to remain budget friendly. The deployment is also limited to platforms like Render (frontend) and Heroku (backend), which have free-tier features, including limited uptime, slow cold starts and bandwidth limits, that limit performance and reliability.

Ethical and Regulatory Obligations: As the system can be related to personal memories, images, and voice records of elderly users, the design has to correspond to the general principles of data protection. Despite the fact that the university does not have a specific legal framework in place, the system is supposed to be based on some basic privacy principles: the media files should be stored in a secure manner, the communication between the API should be encrypted, the sensitive data should be anonymized, and the access to the system should be limited to authenticated sessions. Information related to older people is sensitive, and the system should thus not store unwanted personal information and must have additional secure communication measures (HTTPS, JWT authentication, encrypted credentials storage).

Security/privacy limitations: Security requirements reduce flexibility of design. The frontend, backend, and AI modules should communicate with each other through secure communication. The control of user sessions must be done by JWT or token-based authentication. Since externally provided APIs manipulate images and audio files provided by users, the system should have a secure transmission of these files, and it should not accept harmful and unreliable data. Moreover, memory data should not be accessible to developers; they should not use local storage that can be easily hacked, plaintext logging, or open public endpoints. Such limitations affect the choices of an architect and minimize the application of unsecured lightweight protocols.

Interface and Integration Constraints: The system will need to communicate with the following services: a database service based on PostgreSQL, a media storage service based on Firebase/Cloudinary, service-based media processing, and a deployment service based on hosting. All these systems have integration standards and constraints. As an example, the base 64 encoding of images is needed by certain AI APIs, whereas definite audio formats (WAV/MP3) are required by others. Cloudinary limits file size, whereas PostgreSQL schema designs are faced with the task of managing structured event metadata. The system architecture should thus abide by the constraints of every external interface, tying up the freedom to include personal data formats or non-standard communication protocols.

Performance and Time Constraints: Response times are affected by network connectivity and API latency because of the use of cloud AI services. Image restoration and long-form story generation can be performed in a few seconds to be of high quality. The system should have loading indicators and should not take long waiting periods since the target users are elderly people, which limits the complexity of AI transformations. On the same note, free-tier deployment providers can cause server cold starts, which raises initial loading times. Such timing limitations diminish the possibility of processing, or real-time or near-real-time processing.

Maintenance of Constraints and Process of Development: The infrastructure and the academic calendar in the university provide constraints in terms of time. Usage of GPU labs is time-based, and development has to be in accordance with deadlines for academic projects. The development team should adhere to the usual software engineering concepts taught at UCP, such as design support by SRS, modular architecture, and documented workflow. The university or later student teams might do the maintenance of the system in future, but it would be necessary to follow the clean code practices, version control standards, and documentations. Such limitations minimize unconventional structures or experimental development methods.

Budgetary Constraints: The project is run on a very strict low-cost model with the overall cost projected to be PKR 20,000 to 30,000. This budget will limit the paid AI services, advanced deployment servers, or tools of premium level. The system should utilize the free-tier services and facilities offered by universities as much as possible and should not utilize the high-volume API calls or scaled enterprise hosting services. Consequently, the amount of data that media can store, how often AI works, and how many users are supported at the same time is also limited in nature.

2.5 Assumptions and Dependencies

The remarkable creation and functioning of the ReMIND system are based on various assumptions that state the availability of technologies, institutional assistance, development terms, outer API services, and the condition of the working environment. These are not facts, and they might affect the fullness or accuracy of the requirements. Also, the project relies on various external elements and third-party services, which might affect the performance of the system or its viability in the case of failure, alterations, or any other consequence.

1. Assumptions

The assumption is that the university will keep availing access to high-performance GPU lab machines with NVIDIA RTX GPUs during the project time. The access will be necessary to test and validate AI processing operations like image restoration, model inference, and audio transcription. In case of access, AI processing speed and the ability to test can be considerably decreased.

It is presumed that the personal laptops of team members are working and have the capacity to support local development work including running React, Flask/Fastapi servers, and their ability to connect to remote services. Hardware failures or poor performance than expected may slow down the development or productivity.

The project presupposes that there is a stable internet connectivity to be used during the development and operation of the project. As AI calls, cloud storage and deployment platforms run on the basis of network communication, intermittent network connections can lead to failure or slowdown of the main functionality of memory upload, story generation, and voice transcription.

It is presumed that the free-tier or more academic-tier API credits offered by such services as OpenAI, Whisper, Real-ESRGAN, Google Vision, or ElevenLabs will not expire during the project. A rise in prices, rate limits, or access policies might raise the cost or limit the system's capabilities.

The system presupposes that old users or caregivers are going to be capable of making relatively well-structured inputs (text, voice, or images) which can be processed by AI models. Very ambiguous, noisy, or unclear inputs can decrease the precision of AI-generated outputs. There is also an assumption that a user has to work with the system in a normal home or care setting whereby mobile/desktop devices are available.

It is presumed that the system being made is for those people who have caregivers or someone around who understands the technology and can access the system on a patient's behalf and cooperate with the terms & conditions and privacy policies.

It is supposed that during the time of the development, React.js, Flask/FastAPI, PostgreSQL, Firebase/Cloudinary, and third-party AI APIs will be backward compatible. Other important changes or depreciations in such tools can necessitate component redesign or refactoring.

It is presupposed that the project team will use the necessary software tools like GitHub, Jira, Figma, and Postman on an ongoing basis. The development of coordination, version, or design workflow may be impacted by temporary or permanent restrictions on these tools.

It is assumed proof of concept project, which might be helpful for future works. Any inconvenience due to limitations and lack of resources will be tolerated.

2. Dependencies

The system requires outside AI services to carry out important tasks, such as story creation, image repair, and speech recognition. In case of outages in these services, longer latency, API alterations, or abandonment, the main functionality of ReMIND might not work or stop functioning.

The project will rely on PostgreSQL to store user data and structured memory and Firebase/Cloudinary to store media files. Reconstructed memories could be influenced by any modifications in the database connectivity, storage capacity, or reliability of the service used in the process of retrieving data, media access, and interaction among the users.

System implementation is based on Render and Heroku as frontend and backend respectively. The limitations of free-tier performance enforced by these platforms include uptime limits, cold starts, and bandwidth limits. In case these services alter their free usage, policies or are unavailable due to downtime, the availability and responsiveness of the system can be affected.

The system relies on the lab infrastructure and library resources along with the institutional support of the technical references, research materials, and access to GPUs of the UCP. Alternations in the university policies, labs, or availability of resources can impact the timelines and capabilities of the project to meet the AI workloads.

The system is based on the standard communication protocols (HTTPS, REST APIs) and presupposes their accessibility in devices and networks where elderly users and their caregivers make use of it. Any network firewall or device limitations will interfere with service communication.

The project relies on the capability of the team to adhere to appropriate security procedures, such as storing credentials in a safe and sound manner, using authentication tokens properly, and ensuring the media belonging to the user are handled in a responsible manner. The inability to satisfy these requirements can cause operational risks or vulnerability.

Lastly, the system will be indirectly reliant on users possessing access to compatible devices, i.e. laptops, tablets, or smartphones that are capable of using modern browsers. The device might be incompatible or fail to provide the necessary features (e.g. microphone permission, camera permission), which would restrict the use or eliminate some methods of memory input.

3. Technical Architecture

This section describes the overall technical architecture of the ReMIND system, including major subsystems, processing responsibilities, interfaces, data components, technology stack, and hosting considerations. The architecture incorporates both the operational prototype and the envisioned full-system interactions.

ReMIND is a **custom-built application that leverages Commercial Off-The-Shelf (COTS) AI services**, primarily inspired by and built around the **ChatGPT-style generative AI architecture**. Rather than developing AI models from scratch, the system integrates mature third-party AI APIs for language understanding, image enhancement, speech transcription, and voice synthesis. This approach ensures technical feasibility within constraints while maintaining functional depth.

At a high level, ReMIND follows a **layered, client-server architecture** that supports **online, near real-time processing**. The system handles event-driven transactions such as multimedia uploads, AI processing requests, memory reconstruction, and reminder generation. Analytical reporting is limited to basic logs and feedback analysis, consistent with the proof-of-concept nature of the project.

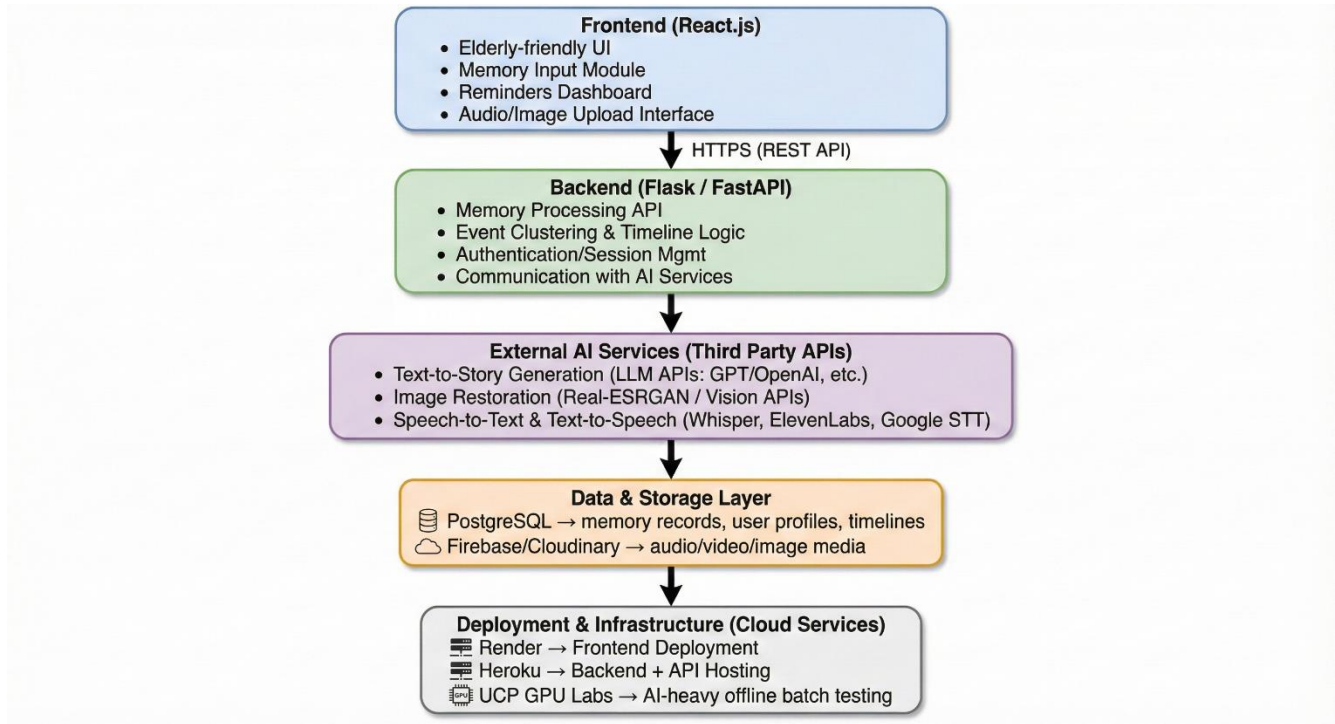


Fig 3.1: Architecture Diagram

Major Application Components and Responsibilities

The **front-end layer** provides a browser-based user interface optimized for elderly users. It enables multimedia input, reconstructed memory visualization, reminder interaction, and feedback submission.

The **backend layer** acts as the core control unit. It manages authentication, workflow orchestration, API communication, and business logic. It coordinates AI requests, validates outputs, and ensures secure storage and retrieval of data.

The **AI orchestration module** serves as a mediator between the backend and external AI services. It aggregates multimodal outputs and initiates memory reconstruction via large language models.

The **data layer** manages structured and unstructured data, including user profiles, memory timelines, reminders, and media references.

3.1 Application and Data Architecture

Component Diagram:

This Component Diagram illustrates the physical architecture of the ReMIND system, showing how the software components interact. The core communication hub is the API Gateway, which routes requests between different modules. The Frontend (React.js) provides user interfaces like the Memory Input UI and Reconstructed Memories Viewer. The Backend (Flask / FastAPI) handles User Management and Event Clustering. The system relies on external AI Services for tasks like Story Generation and Speech-to-Text. Data is persistently stored in a PostgreSQL Database and Cloudinary / Firebase Media Storage. The Notification Module handles reminders, and the entire system is deployed using Render and Heroku.

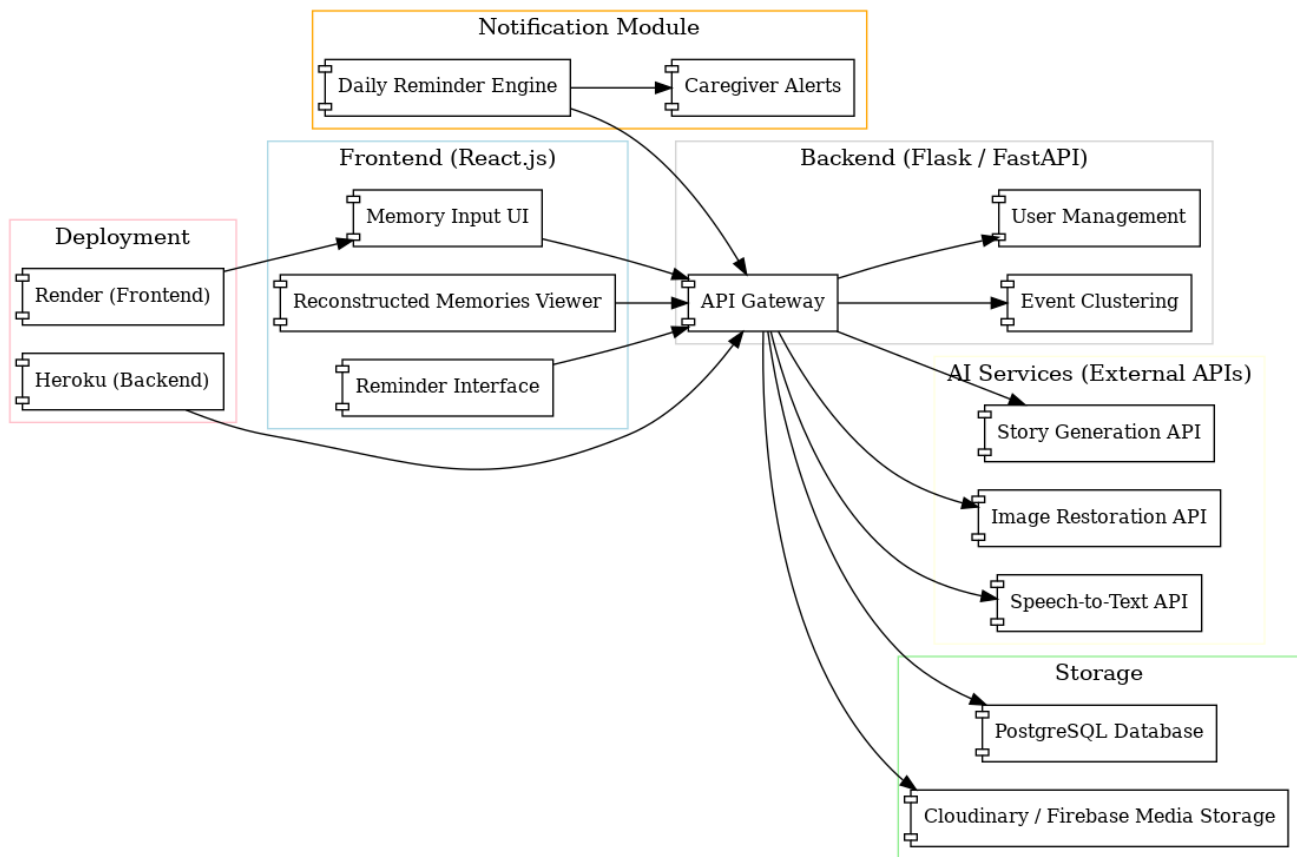


Fig 3.1.1: Component Diagram

ERD Diagram:

This Entity-Relationship Diagram (ERD) visualizes the structure of a database for ReMIND. It models key entities such as the adaptable User (specialized as Patient, Caregiver, and Admin) and the relationships that connect them. Patients, associated with Conditions, receive care and are managed by Caregivers. The operational core includes Chat Sessions for communication, composed of Messages and Media Files, with support for Reminders (which issue Notifications), and the capture of user Feedback. All critical activities are logged in an Audit Log, and the system includes functionality for processing media using AI Processing Tasks.

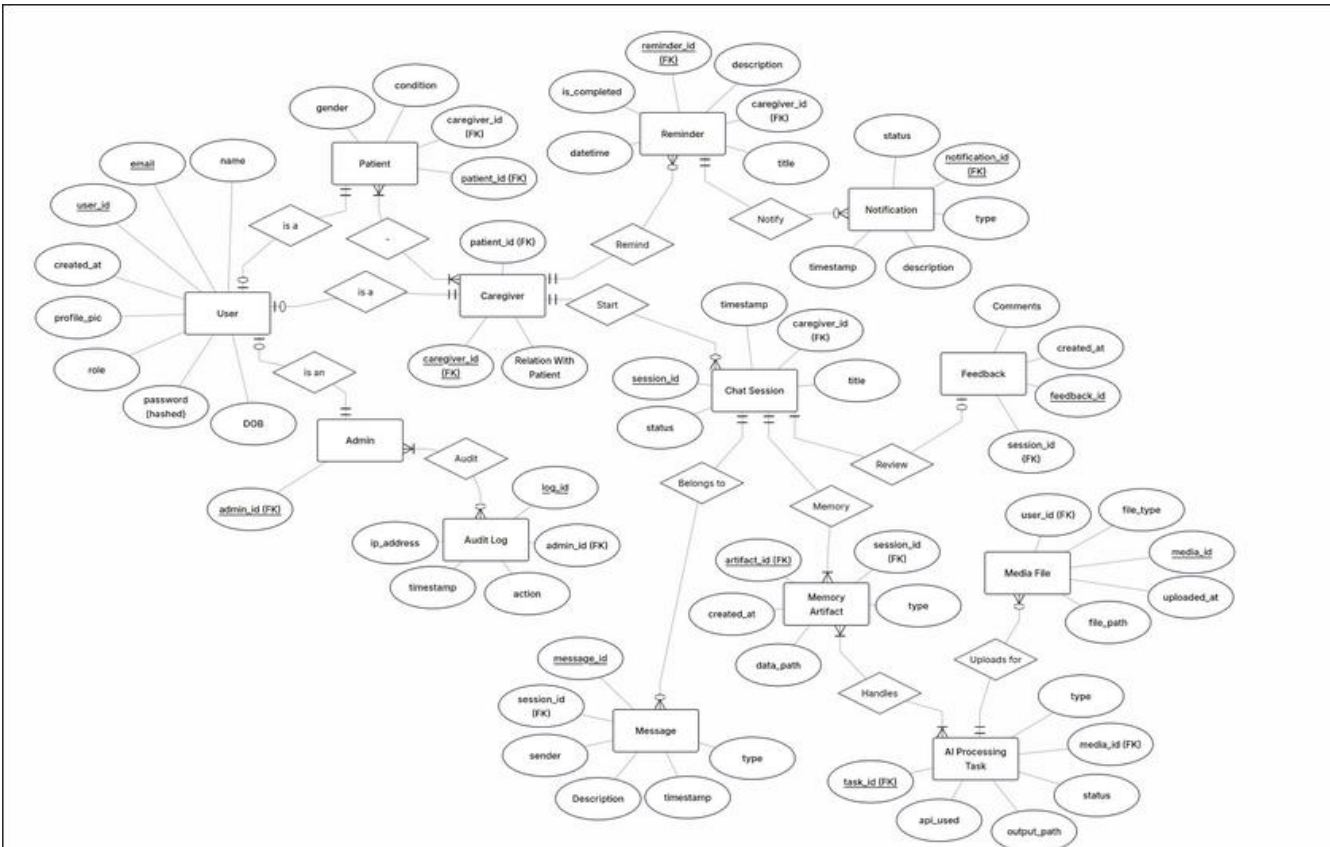


Fig 3.1.2: ERD Diagram

UML Class Diagram:

This UML Class Diagram models the structure of ReMIND. It organizes functionality into modules: Profiles, Audit Log, Caregiver & Patient, Chat, and AI & Media. The Profiles module defines User, specialized in Caregiver Profile, Patient Profile, and Admin Profile. The Caregiver & Patient Module manages Reminders and Notifications. The Chat Module handles Chat Sessions, Messages, and Feedback. Finally, the AI & Media Module defines Media, Memory Artifact, and AI Processing Task for handling media content.

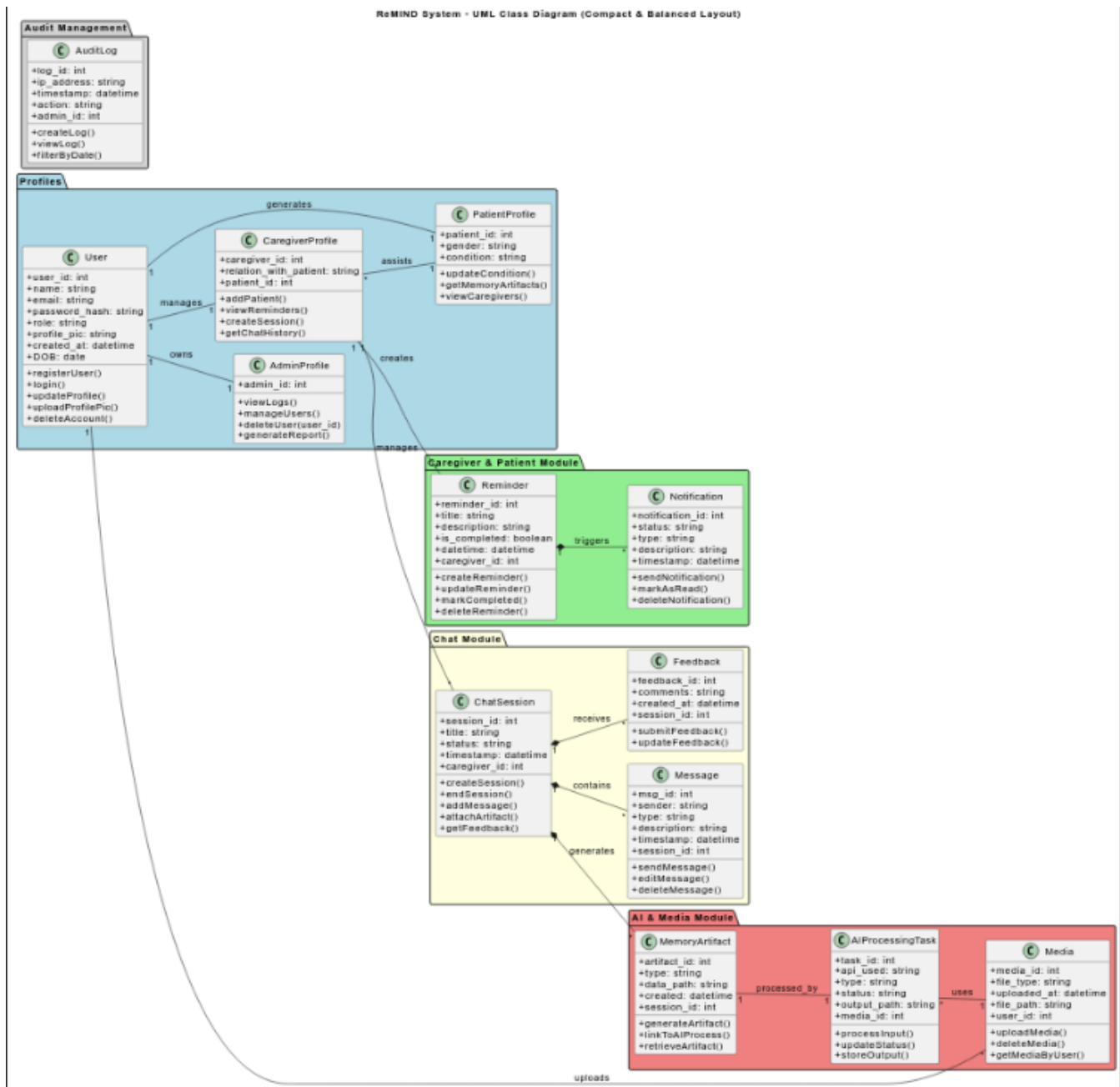


Fig 3.1.3: UML Class Diagram

Relational Schema:

This relational schema, based on the ReMIND-DATABASE ERD. It features a central User table, with specialized child tables: AdminProfile, CaregiverProfile, and PatientProfile, linked by foreign keys (FK) on the respective ID fields. The core interactions occur in the ChatSession table, which references CaregiverProfile and PatientProfile. Communication data is stored in Message and Media tables. Support functionality includes Reminder and Notification tables. All actions are tracked in the Auditlog table, while AIProcessingTask and MemoryArtifact handle media processing and retrieval.

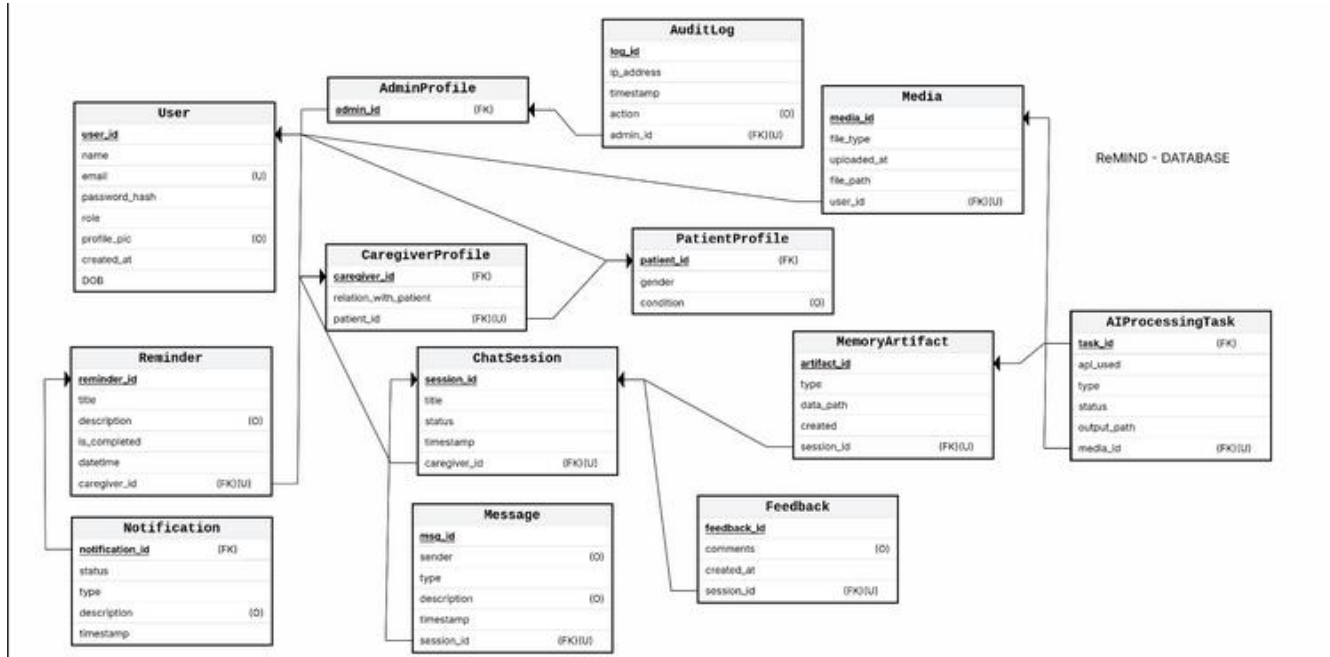


Fig 3.1.4: Relational Schema

Decision Table:

Rule	Input	Alternative Conditions	Action	Output
R1	User provides valid login credentials	Invalid credentials	Authenticate user, create session	Access granted / error message displayed
R2	Patient uploads multimedia input	Unsupported format or missing metadata	Accept input, store temporarily, trigger preprocessing	Acknowledgment / error message
R3	Uploaded file validated successfully	File corrupted or exceeds size limit	Proceed to AI preprocessing and enhancement	Process initiated / warning prompt
R4	Image input detected	No image detected	Trigger Real-ESRGAN enhancement and tagging	Enhanced image + metadata stored
R5	Audio input detected	No audio provided	Send to Whisper API for transcription	Transcript text returned
R6	Text or transcript available	Missing or empty text input	Send to GPT-4 for contextual understanding	Structured text/context response

ReMIND: Generative AI based Memory Rebuilder

R7	All media processed successfully	Any AI module fails	Aggregate data and reconstruct event	Reconstructed memory / retry message
R8	Narrative generation successful	AI fails to generate or timeout	Store reconstructed memory in repository	Text + voice response / system retry
R9	Reminder set by caregiver	Reminder missing or invalid	Save reminder in database	Reminder confirmation / validation error
R10	Feedback submitted by patient/caregiver	Feedback skipped	Store feedback in log	Confirmation message / skip noted
R11	Admin monitors system logs	Logs not accessible	Display system status dashboard	Log summary / permission error

3.2 Component Interactions and Collaborations

Design Sequence Diagram:

The process flow begins with User Authentication, where the Frontend UI communicates with the Auth Service to validate credentials. Following login, the diagram maps the lifecycle of a memory: the Frontend sends data to the Backend (Flask/FastAPI), which coordinates with Cloud Storage (Cloudinary) for media and triggers AI Services (like Whisper API for speech-to-text or GPT-4 for analysis). These services process the data and return it to the Memory Reconstruction Engine. Finally, the rebuilt memory is saved to the Database and delivered back to the Frontend for the user to view.

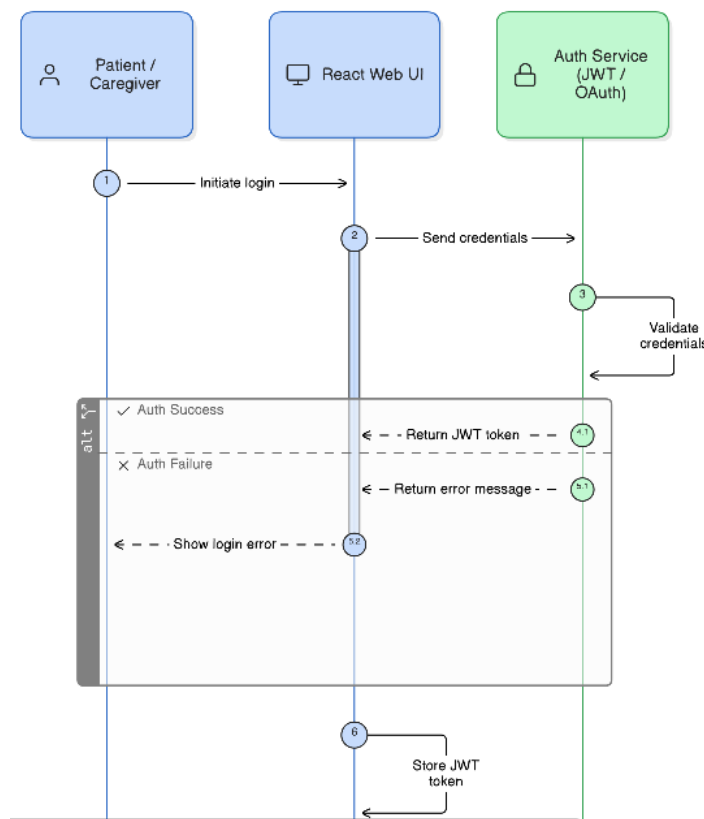


Fig 3.2.1: SD1

ReMIND: Generative AI based Memory Rebuilder

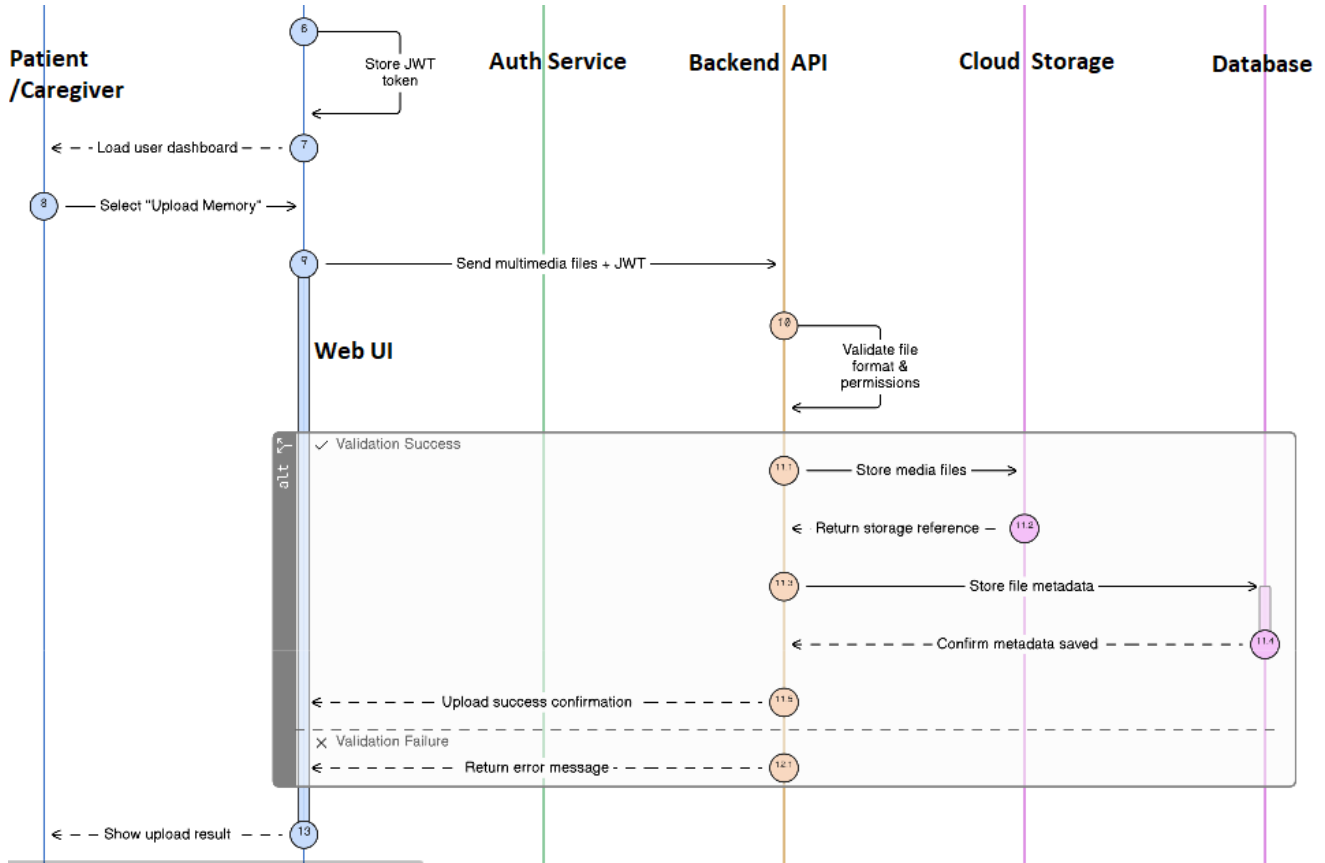


Fig 3.2.2: SD2

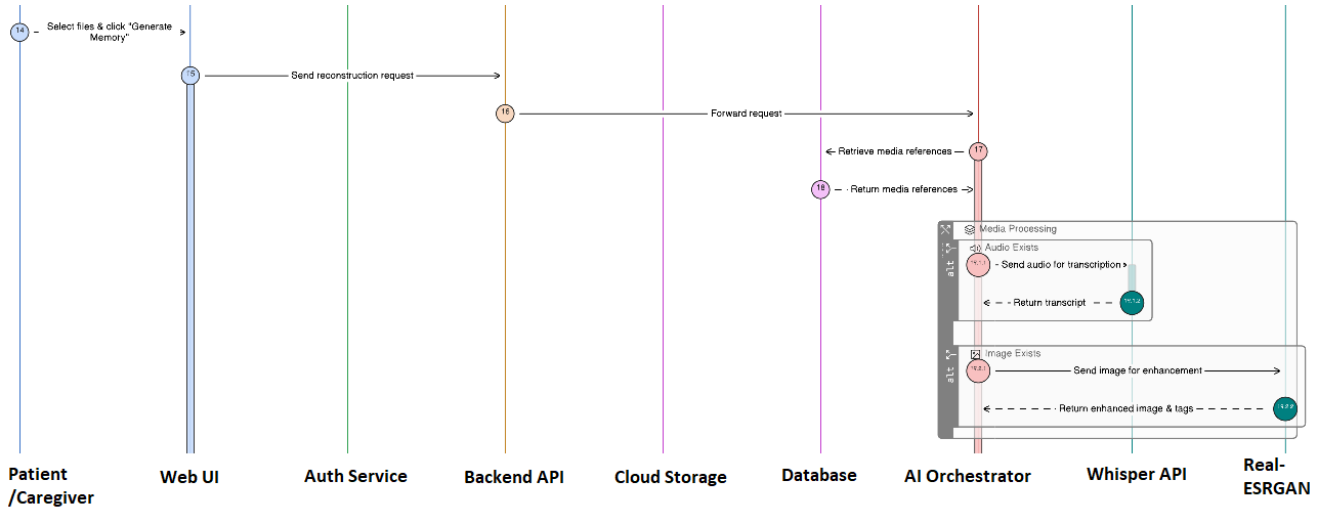


Fig 3.2.3: SD3

ReMIND: Generative AI based Memory Rebuilder

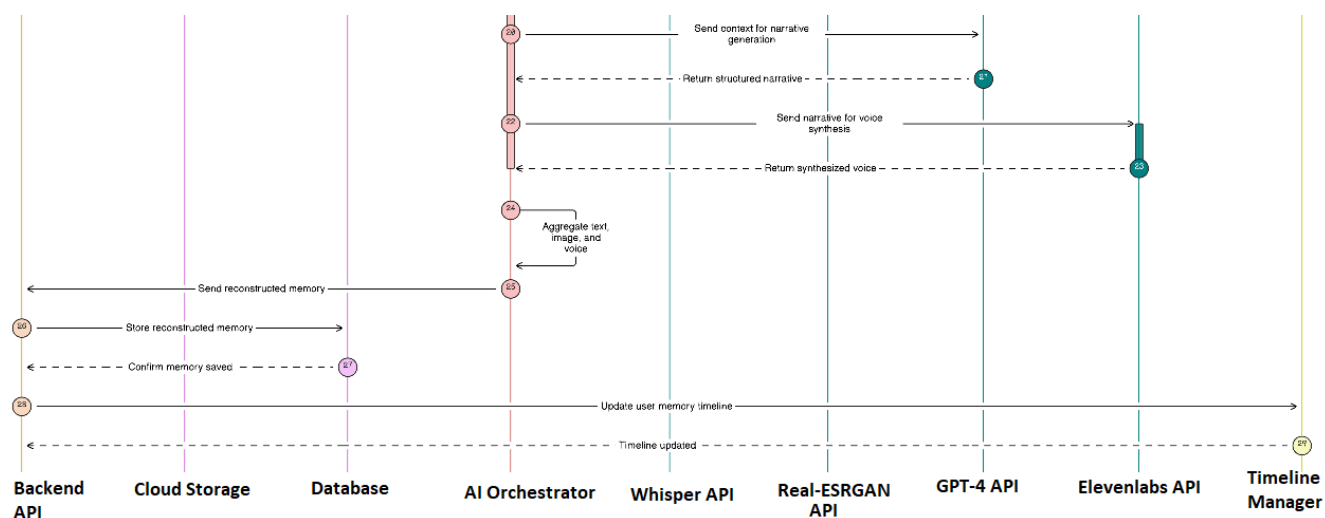


Fig 3.2.4: SD4

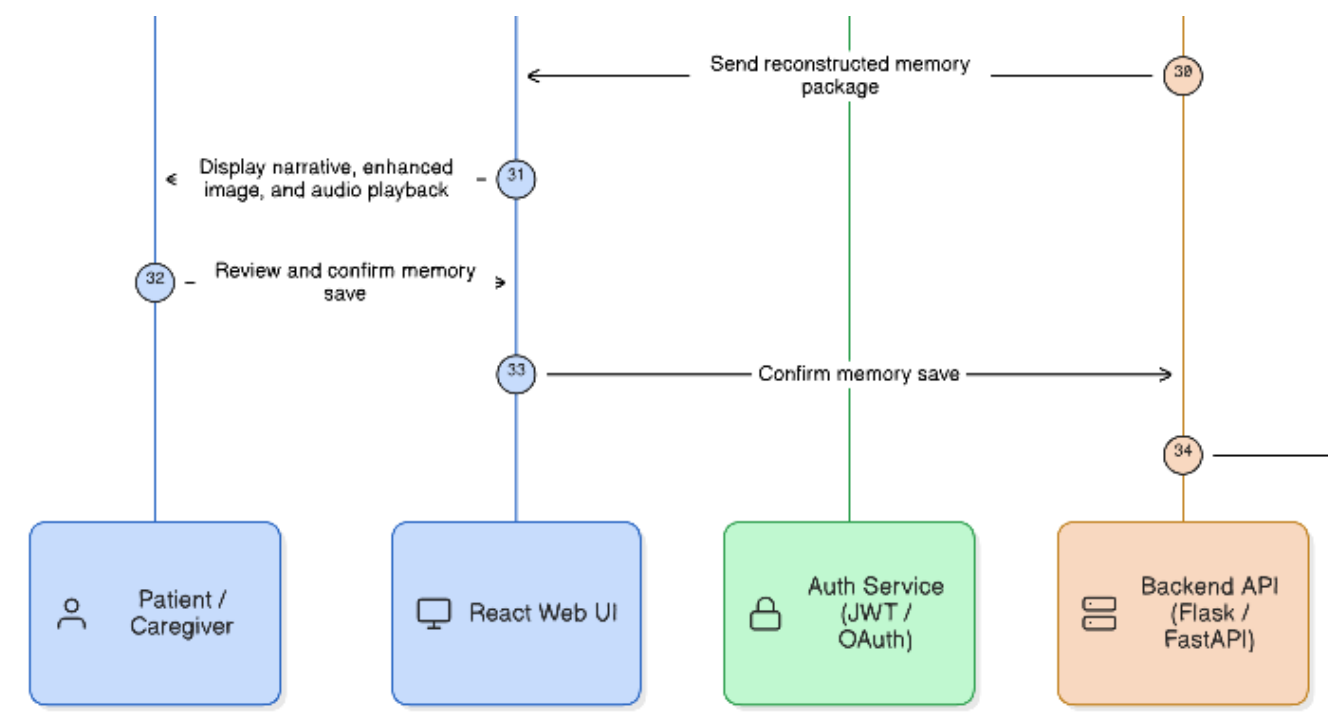


Fig 3.2.5: SD5

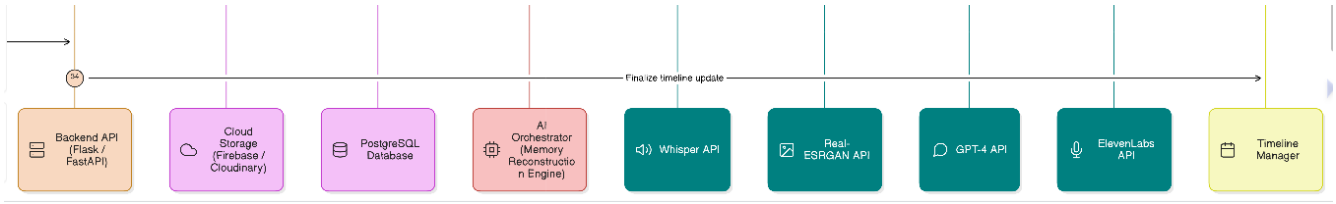


Fig 3.2.6: SD6

Collaboration Diagram:

This Collaboration Diagram illustrates the sequence of messages exchanged between the three main logical components of the ReMIND system: the User / Frontend (React.js), the Backend (Flask / FastAPI), and the AI + Database (APIs + PostgreSQL + Storage). The memory processing workflow starts when the Frontend submits a Memory Input (1) to the Backend. The Backend may then request Additional Details (2) from the Frontend before submitting the text/media for AI Processing (3) to the AI + Database component. After processing, the AI + Database component returns the AI-Generated Output (4) to the Backend and confirms Storage Confirmation (6) after saving the memory and media (5). The Backend then delivers the Rebuilt Memory (7) back to the Frontend. Independently, a Reminder Scheduled (8) event from the Frontend triggers the system to Send Reminder Notification (9) via the AI + Database component.

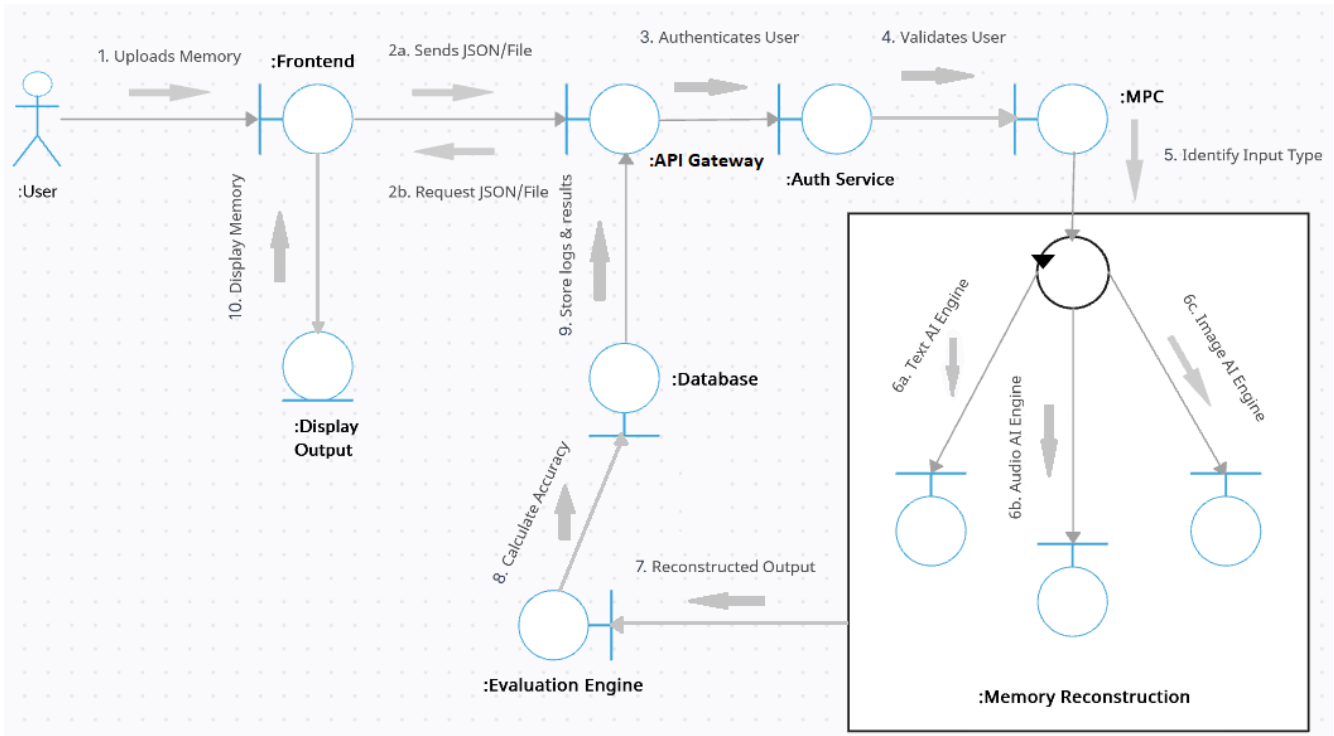


Fig 3.2.7: Collaboration Diagram

Activity Diagram:

This Activity Diagram models the sequential flow of actions a user takes within the ReMIND system, primarily focusing on memory handling. The process begins with Start and proceeds to Login. After successful login, the user must answer Security Questions (a combination of multiple-choice and input) for verification. Next, the user initiates Memory Input/Upload, likely submitting multimedia content or textual data. This input is then subjected to Memory Analysis/Processing, which transforms the raw data. The processed memory is then handled in Memory Storage & Tagging, where it is saved and categorized. Finally, the system executes Memory Retrieval/Display, presenting the reconstructed or stored memories to the user before the process reaches End.

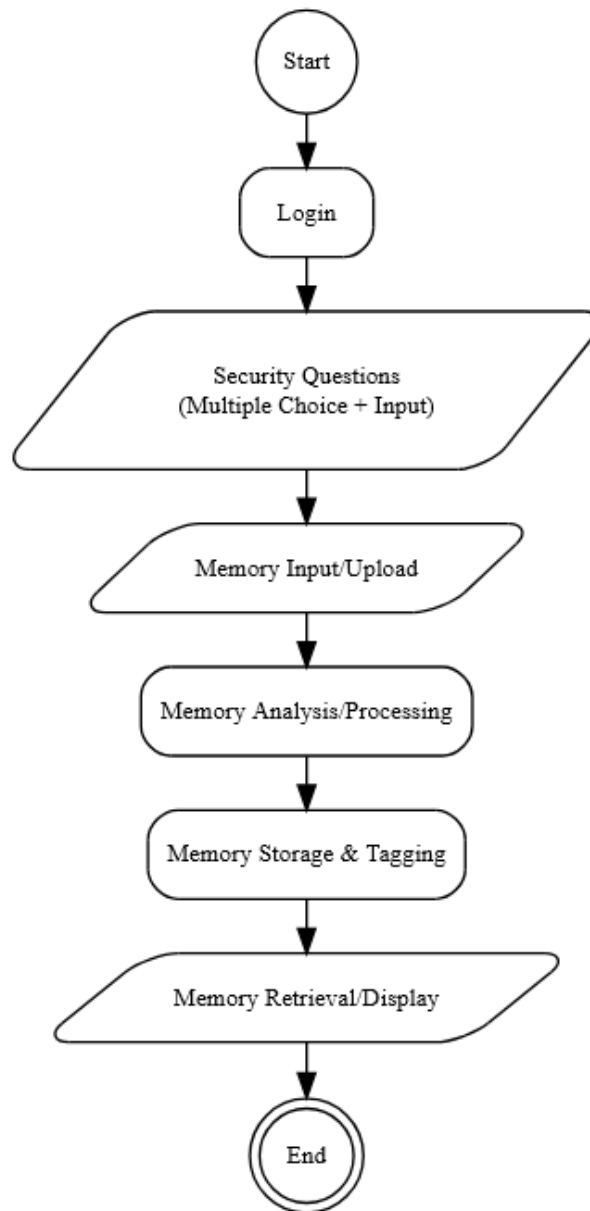


Fig 3.2.8: Activity Diagram

Event Table:

Event	Event Name	Trigger Type	Source	System Activity	Output
E1	User Login	External (User Action)	Patient / Caregiver / Admin	Authenticate credentials, create session	Dashboard or role-based access
E2	Media Upload	External (User Action)	Patient / Caregiver	Validate and store uploaded media	Temporary file storage + confirmation
E3	Data Preprocessing	Internal (System)	Backend	Cleans, formats, validates input data	Ready media for AI modules
E4	Image Enhancement	External (AI Service)	Real-ESRGAN	Enhances image and extracts context	Enhanced image file stored
E5	Audio Transcription	External (AI Service)	Whisper API	Converts audio to text, detects tone	Transcript returned
E6	Text Analysis	External (AI Service)	GPT-4	Interprets and structures text context	Semantic data created
E7	Event Reconstruction	Internal (AI Process)	AI Module	Combines multimodal data into cohesive memory	Structured narrative generated
E8	Voice Synthesis	External (AI Service)	ElevenLabs	Converts narrative text to natural audio	Voice file generated and stored
E9	Memory Storage	Internal (Database)	Backend → PostgreSQL	Saves reconstructed memory with metadata	Record created in memory repository
E10	Memory Retrieval	External (User Request)	Patient / Caregiver	Fetches requested memory or timeline	Displayed in frontend interface
E11	Reminder Creation	External (User Action)	Caregiver	Adds reminder entry in database	Confirmation sent to user
E12	Reminder Notification	Internal (Timed Event)	Notification Module	Sends scheduled alert to patient	Notification on UI
E13	Feedback Submission	External (User Action)	Patient / Caregiver	Logs ratings and suggestions	Stored in feedback log
E14	System Monitoring	Internal (Admin Action)	Administrator	Reviews logs, checks API integrations	Diagnostic report generated

Detailed DFD with Explanation:

DFD Level 1:

The DFD Level 1 diagram details the three main high-level processes that govern the system's operation: 3.0 User Authentication, 1.0 Data Processing, and 2.0 Memory Reconstruction. User Authentication (3.0) is the entry point, handling login/registration info from the Patient, Caregiver, and Admin entities and storing/retrieving user data from the Accounts DB. It also grants access permissions to Data Processing (1.0) and forwards session validation to Memory Reconstruction (2.0). Data Processing (1.0) receives multimedia uploads from the Patient and Caregiver; stores validated data in the Data Process DB and forwards the processed data to Memory Reconstruction (2.0). Finally, Memory Reconstruction (2.0) generates Memory Results/Feedback for the users, sends data analysis to AI Services, stores reconstructed memories in the Memory Artifacts DB, and provides reports/statistics to the Admin.

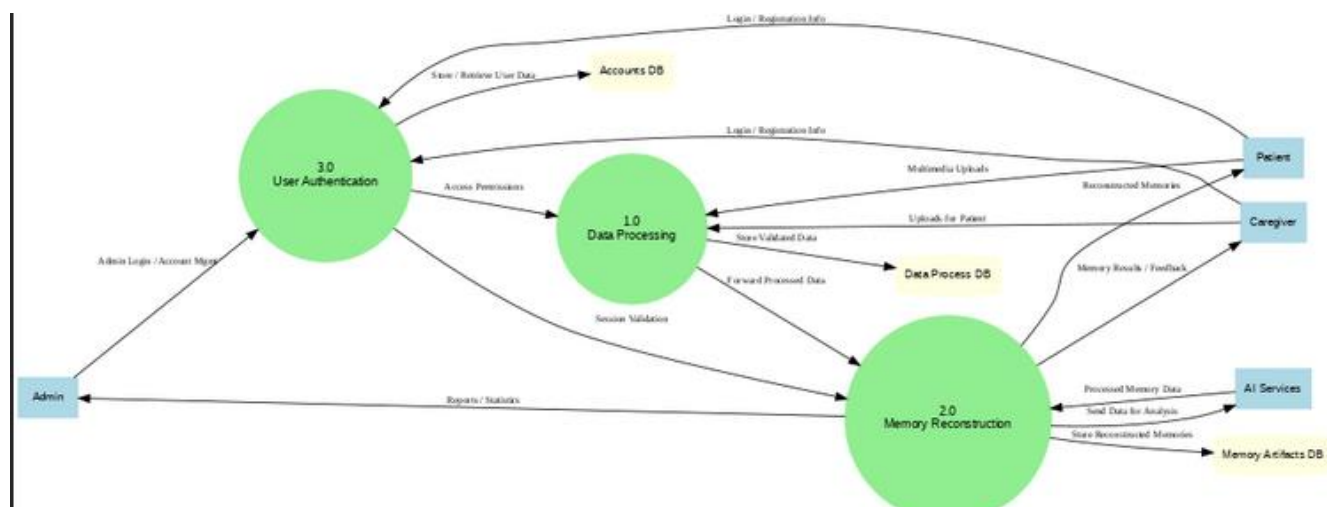


Fig 3.2.9: DFD Level 1

DFD Level 2:

The DFD Level 2 diagram elaborates on the three main processes of the system: 3.0 User Authentication, 1.0 Data Processing, and the now-decomposed 2.0 Memory Reconstruction. The Memory Reconstruction process (2.0) is broken down into two sub-processes: 2.0.1 API Calls & Integration and 2.0.2 Memory Verification. Data Processing (1.0) still receives uploads from the Caregiver and Patient and sends data to the Data Process DB, but it now forwards processed data to both the main Memory Reconstruction (2.0) process and 2.0.1 API Calls & Integration. API Calls & Integration (2.0.1) handles sending/receiving media from AI Services, logging API requests/responses in the APIs DB, and interacting with Data Processing (1.0). Finally, Memory Verification (2.0.2) receives data from the main Memory Reconstruction (2.0) process, verifies memories using multimedia from the Patient, stores the results in the Memory Artifacts DB, and sends verification results/feedback to the Admin.

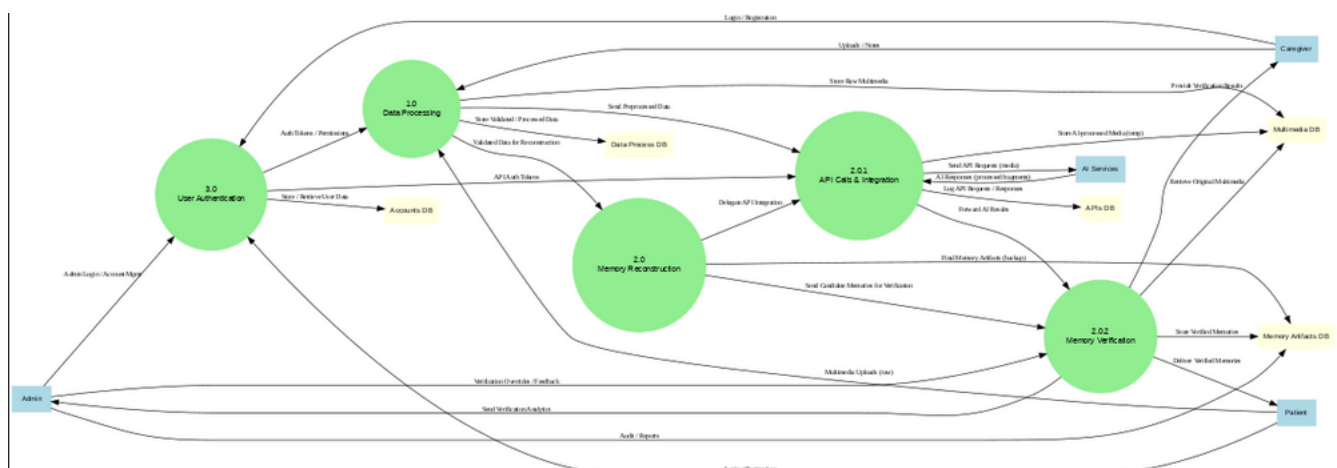


Fig 3.2.10: DFD Level 2

3.3 Design Reuse and Design Patterns

The development of ReMIND heavily leverages software reuse principles to ensure feasibility, reliability, and rapid prototyping within limited academic time and resources. Instead of developing low-level components from scratch, the system integrates externally available, well-tested frameworks, libraries, and commercial off-the-shelf (COTS) services, particularly in the domain of artificial intelligence.

At the application layer, widely adopted open-source frameworks such as React.js (frontend) and Flask/FastAPI (backend) are reused for interface development, API handling, and request orchestration. These frameworks provide standardized routing, state management, and security features, significantly reducing development complexity and risk.

At the intelligence layer, ReMIND reuses pre-trained generative and perceptual AI models offered through commercial APIs, including GPT-4 for narrative generation, Whisper for speech transcription, Real-ESRGAN for image enhancement, and ElevenLabs for voice synthesis. This reuse eliminates the need for custom model design, training, and optimization, making the project technically feasible while ensuring high-quality outputs.

In the data layer, existing database management systems (PostgreSQL) and cloud-based media storage services are reused for persistent storage, backup, and access control.

Additionally, development tools such as GitHub for version control, Postman for API testing, and Figma for UI prototyping are reused as part of standard software engineering practice.

Overall, approximately 30–40% of the system relies on reused components, while 60–70% represents original work focused on system integration, workflow design, data modeling, and user-centric application logic.

3.4 Technology Architecture

Data and Processing Characteristics

ReMIND **collects and manages:**

- User credentials and role information
- Multimedia inputs (text, images, audio)
- AI-generated narratives and voice outputs
- Memory timelines and reminder schedules
- User feedback and system logs

Processing is **online and event-driven**, with asynchronous AI calls to external services. Each user action triggers a transaction that results in AI processing, data persistence, and user-facing output.

Technology Stack and Platforms

Python, JavaScript are **Programming languages** used.

React.js and CSS3 are used to create **Front-end**.

Flask/FastAPI is used to create **Back-end**.

OpenAI GPT-4, Whisper, ElevenLabs, Real-ESRGAN are our **COTS AI Services**.

PostgreSQL is used to **manage databases**.

Cloud-based media storage (Firebase/Cloudinary) is used for **Media storage and management**.

External cloud infrastructure (e.g., Render / cloud servers) for **hosting**.

Personal laptops for development, university GPU systems for AI integrated workloads as **hardware**.

Our system is supported on all browsers on any desktop or mobile device for an **end-user interface**.

Interface and Network Architecture

The system provides a **browser-based user interface** accessible over the **Internet (WAN)**. Communication between frontend, backend, and AI services occurs via secure RESTful APIs. The architecture supports remote access for caregivers and administrators without requiring specialized client software.

4. Screenshots/Prototype

4.1 Workflow

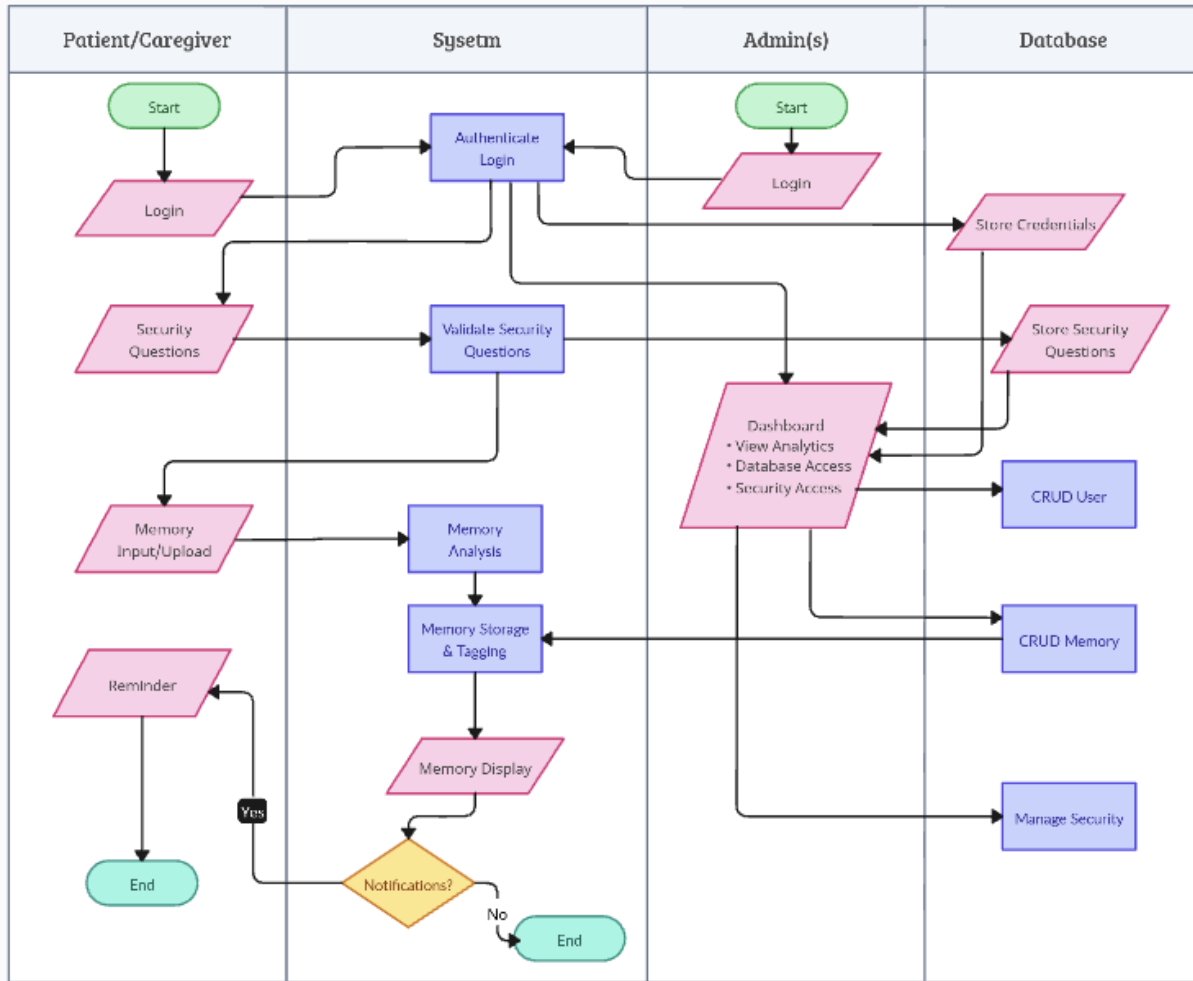


Fig 4.1.1: Swim-Lane Diagram

The workflow of ReMIND begins when a Patient/Caregiver or an Admin initiates the process through a web-based interface. This start phase requires a secure Login, which triggers the System to perform an authentication check. To ensure robust security, the user must then answer Security Questions, which the System validates against records in the Database. This Database layer is central to the operation, securely storing credentials and security question data to facilitate these initial access steps.

Once the user is successfully authenticated, the Patient/Caregiver can provide a Memory Input/Upload consisting of text, images, or audio. The System immediately moves this data into a Memory Analysis phase, where fragmented inputs are processed for contextual meaning. Following analysis, the System performs Memory Storage and Tagging, interacting with the Database to ensure the data is categorized correctly. For the Administrative side, the admin manages the platform via a Dashboard that allows for viewing analytics and performing CRUD operations on both Users and Memories within the Database. The admin also handles the Manage Security function to maintain system integrity.

The final stage of the workflow involves the System generating a Memory Display for the user to review. After the memory is presented, the System checks the status of Notifications. If the user has notifications enabled, the process triggers a Reminder for the Patient/Caregiver to support daily memory reinforcement before reaching the end of the session. If notifications are not enabled, the workflow proceeds directly to the end, concluding the cycle of memory reconstruction and administrative oversight.

4.2 Screens

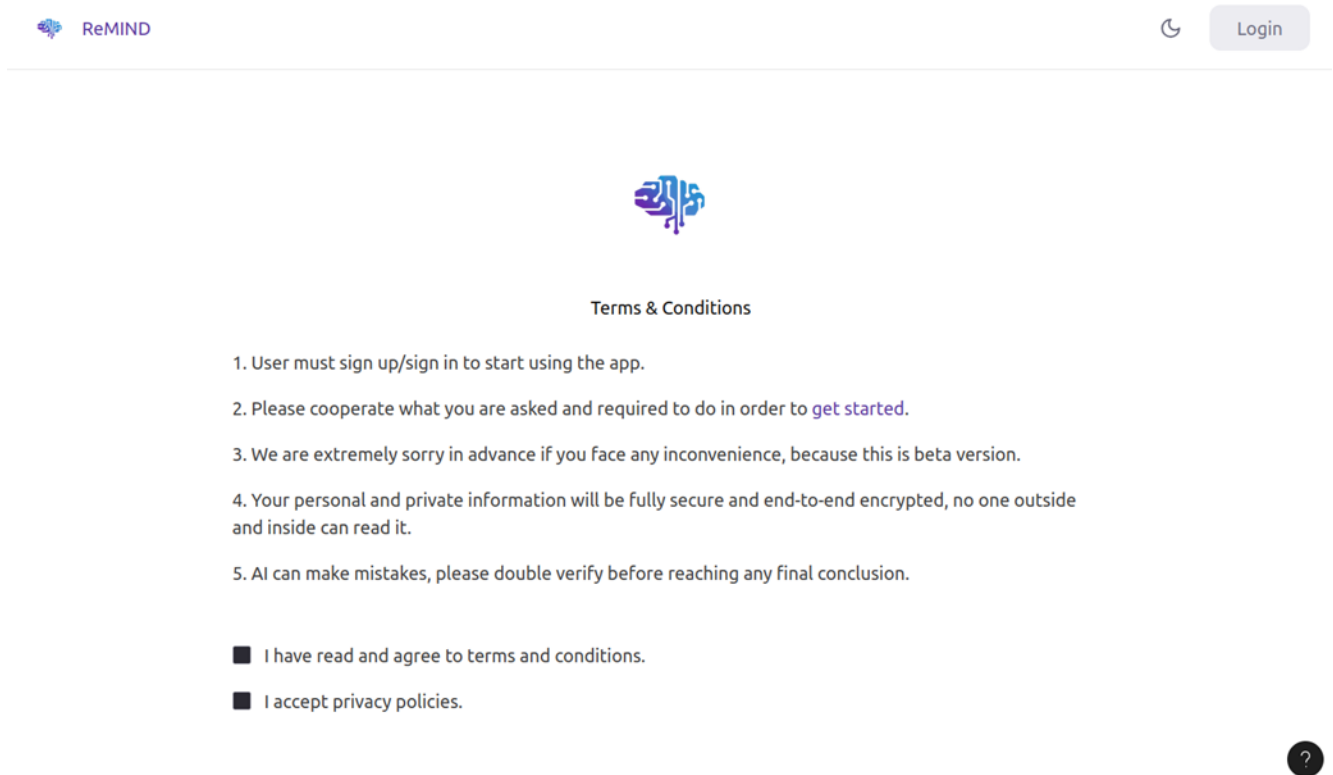


Fig 4.2.1: Logged Out State in Light Mode

ReMIND: Generative AI based Memory Rebuilder

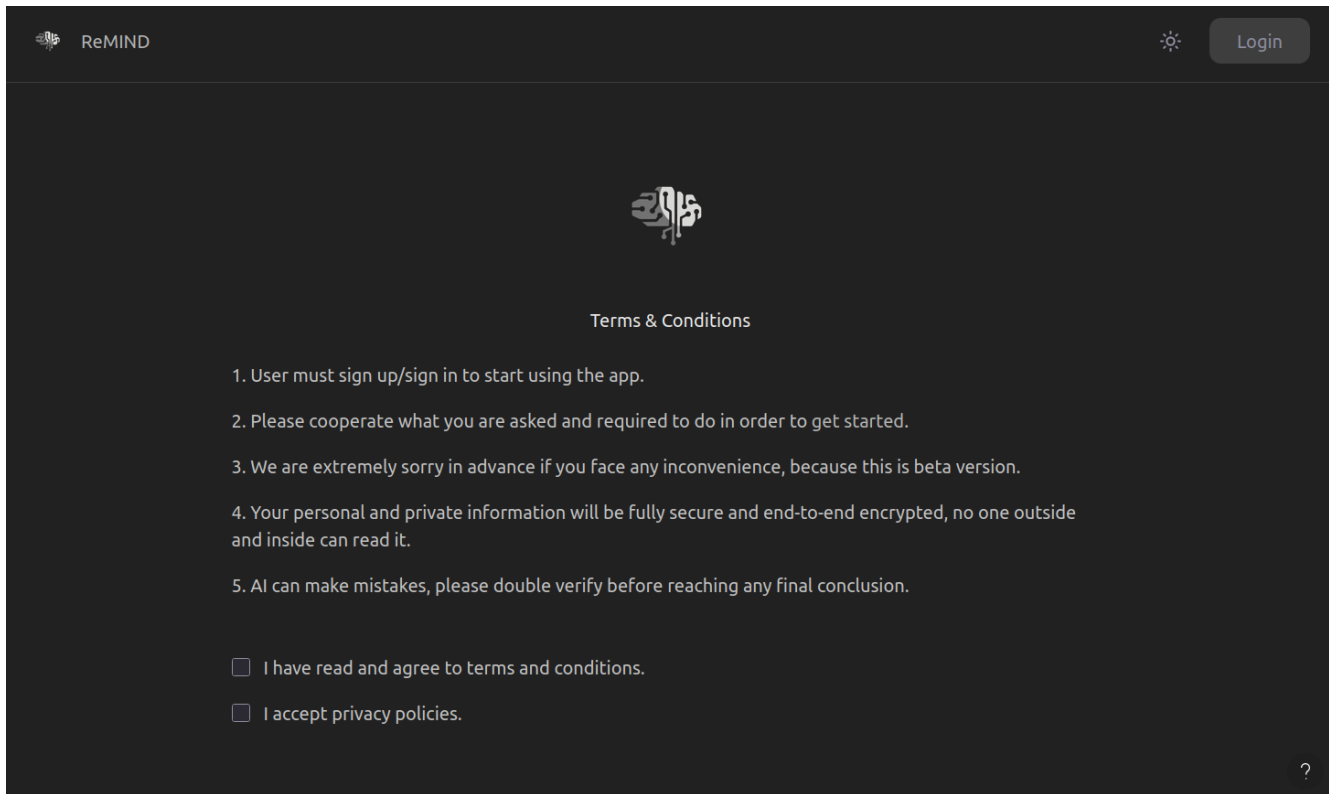


Fig: 4.2.2: Logged Out State in Dark Mode

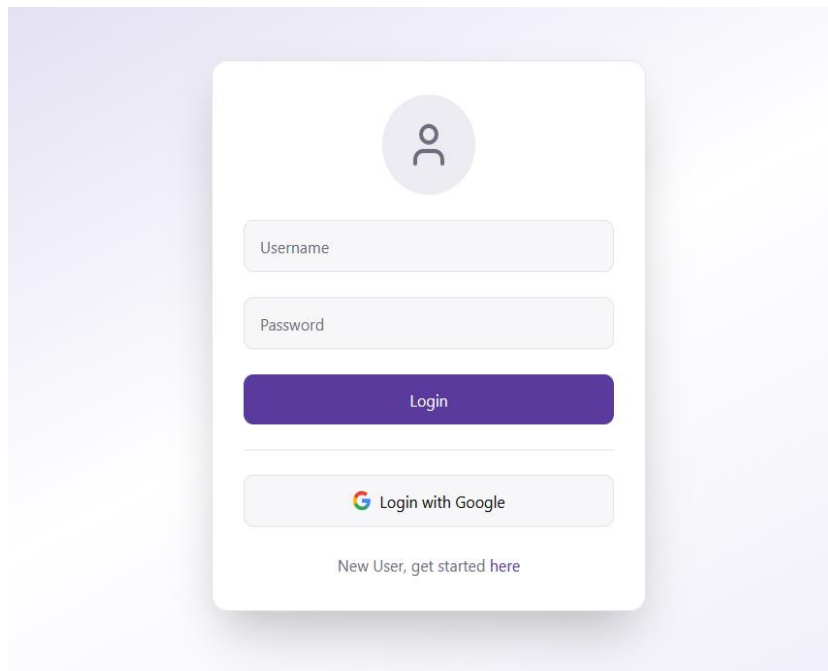
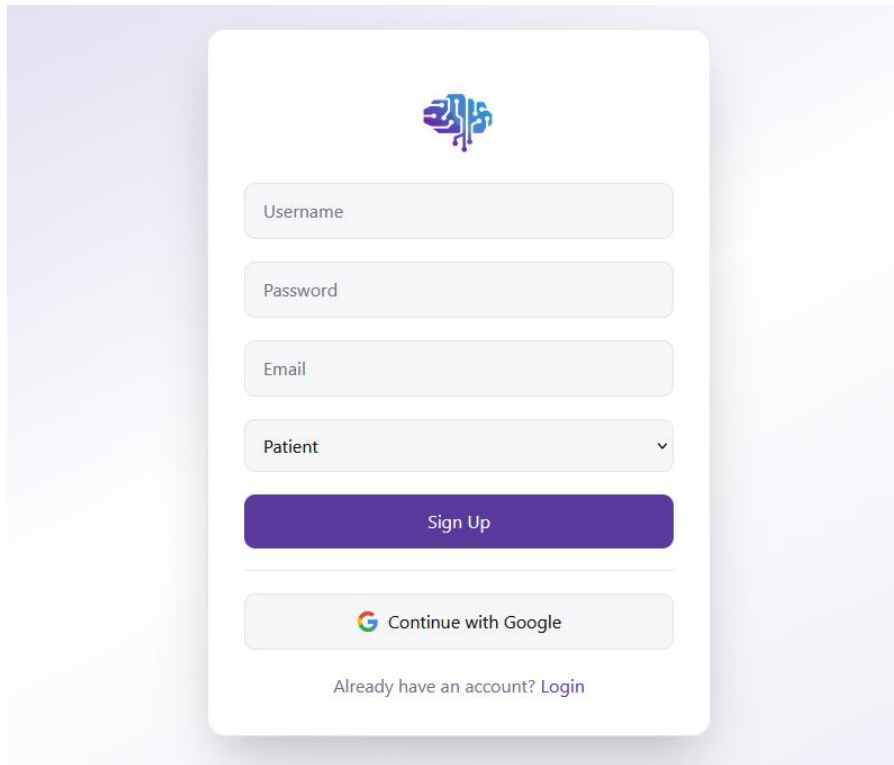
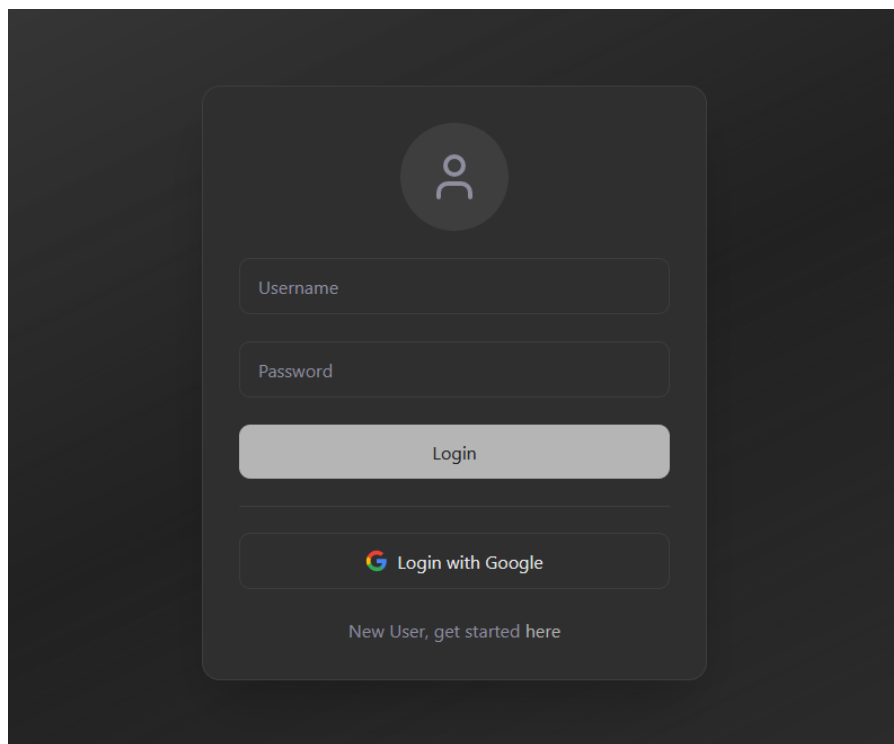


Fig 4.2.3: Login in Light Mode



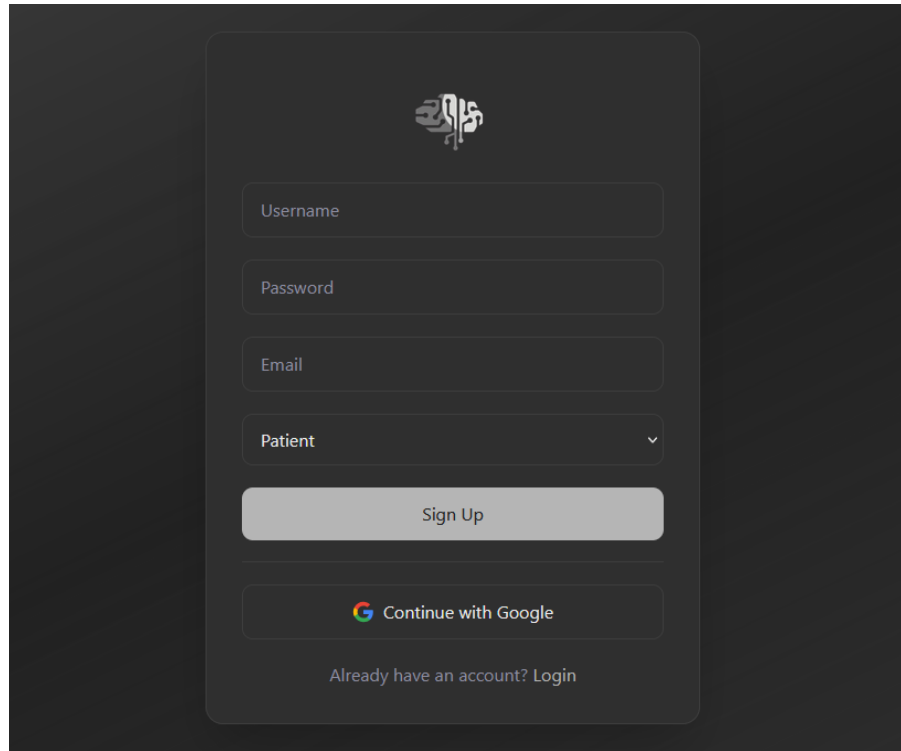
The image shows a sign-up form in light mode. At the top is a logo of a brain with circuit-like patterns. Below it are four input fields: 'Username', 'Password', 'Email', and 'Patient' (which is a dropdown menu). A purple 'Sign Up' button is positioned below the 'Patient' field. A horizontal line separates this from a 'Continue with Google' button, which features the Google logo. At the bottom, there is a link that says 'Already have an account? Login'.

Fig 4.2.4: Sign-up in Light Mode



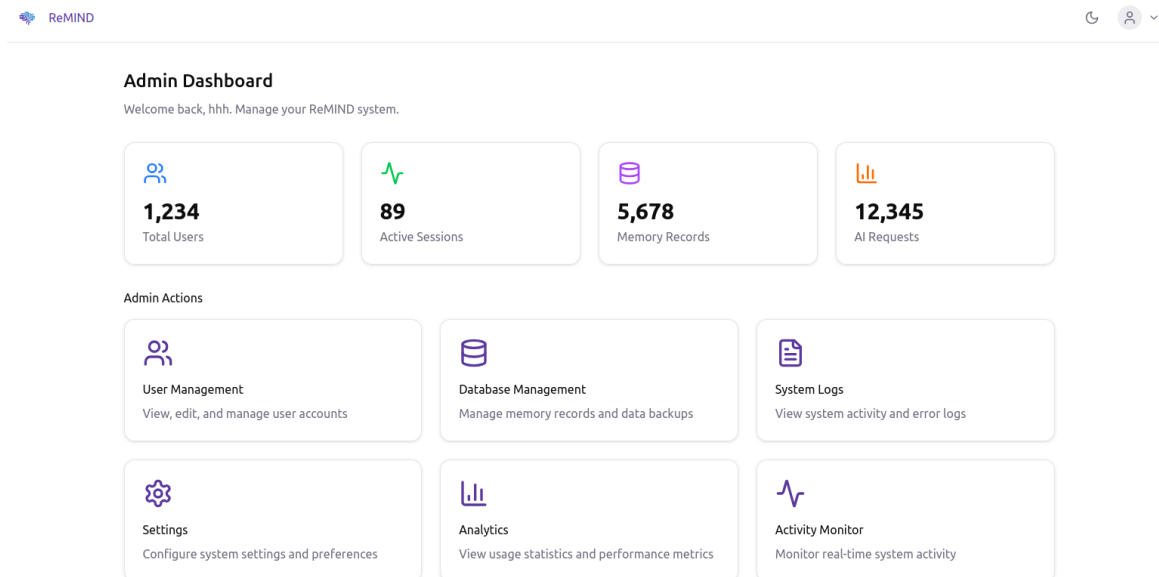
The image shows a login form in dark mode. At the top is a circular icon containing a stylized person silhouette. Below it are two input fields: 'Username' and 'Password'. A light gray 'Login' button is located below the 'Password' field. A horizontal line separates this from a 'Login with Google' button, which features the Google logo. At the bottom, there is a link that says 'New User, get started here'.

Fig 4.2.5: Login in Dark Mode



A dark-themed sign-up form for the ReMIND system. The form is centered on a dark background and contains the following elements: a logo at the top, followed by input fields for Username, Password, and Email. Below these is a dropdown menu labeled 'Patient'. A large 'Sign Up' button is positioned below the dropdown. Underneath the button is a 'Continue with Google' button with the Google logo. At the bottom of the form, there is a link that says 'Already have an account? Login'.

Fig 4.2.6: Sign-up in Dark Mode



The Admin Dashboard for the ReMIND system. It features a header with the ReMIND logo and a user profile icon. The main content area is titled 'Admin Dashboard' and includes a welcome message: 'Welcome back, hhh. Manage your ReMIND system.' Below this, there are four summary cards showing key metrics: Total Users (1,234), Active Sessions (89), Memory Records (5,678), and AI Requests (12,345). Under the 'Admin Actions' section, there are six cards for various management tasks: User Management, Database Management, System Logs, Settings, Analytics, and Activity Monitor, each with a brief description of the action.

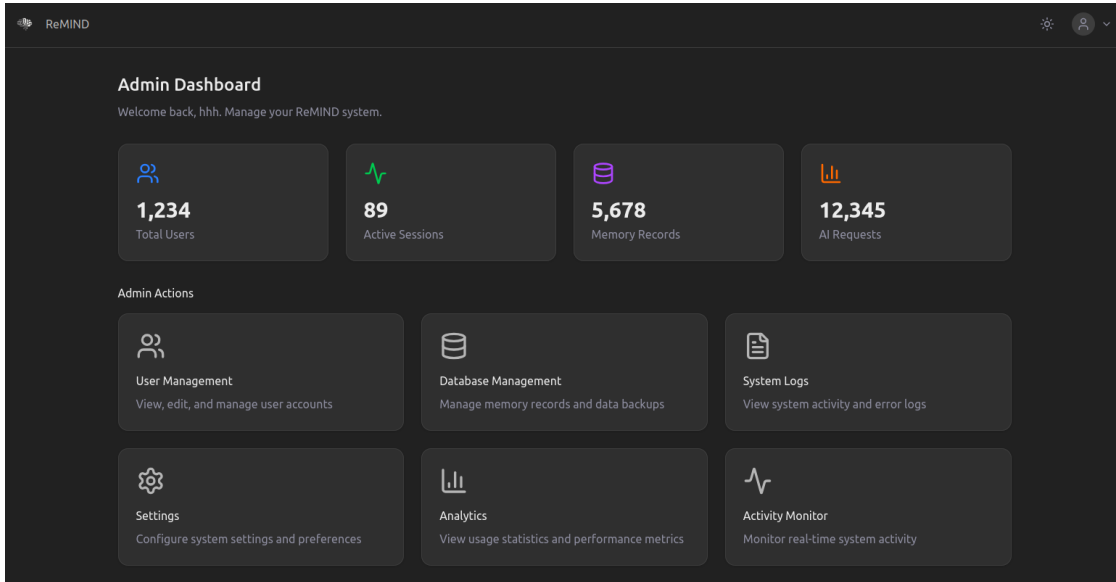
Admin Dashboard			
Welcome back, hhh. Manage your ReMIND system.			
 1,234 Total Users	 89 Active Sessions	 5,678 Memory Records	 12,345 AI Requests
Admin Actions			
 User Management View, edit, and manage user accounts	 Database Management Manage memory records and data backups	 System Logs View system activity and error logs	
 Settings Configure system settings and preferences	 Analytics View usage statistics and performance metrics	 Activity Monitor Monitor real-time system activity	

ReMIND: Generative AI based Memory Rebuilder

Recent Activity

John Doe Created new memory	5 minutes ago
Jane Smith Updated profile	12 minutes ago
Bob Johnson Shared chat	23 minutes ago
Alice Brown Deleted memory	1 hour ago
Charlie Wilson Logged in	2 hours ago

Fig 4.2.7: Admin Dashboard in Light Mode



Recent Activity	
John Doe Created new memory	5 minutes ago
Jane Smith Updated profile	12 minutes ago
Bob Johnson Shared chat	23 minutes ago
Alice Brown Deleted memory	1 hour ago
Charlie Wilson Logged in	2 hours ago

Fig 4.2.8: Admin Dashboard in Dark Mode

ReMIND: Generative AI based Memory Rebuilder

The screenshot shows the ReMIND web application interface. On the left, there is a sidebar with a search bar labeled "Search chats", a "New chat" button, and a message "No chats yet". The main area displays "Question 1 of 5" with a progress indicator. The question is "Is patient male or female?". Below the question are three radio button options: "Male", "Female", and "Prefer not to say".

ReMIND

Search chats

New chat

No chats yet

Question 1 of 5

Is patient male or female?

Male

Female

Prefer not to say

Fig 4.2.9: Security Question 1

The screenshot shows the ReMIND web application interface. On the left, there is a sidebar with a search bar labeled "Search chats", a "New chat" button, and a message "No chats yet". The main area displays "Question 2 of 5" with a progress indicator. The question is "What is age of patient?". Below the question are five radio button options: "0-18", "19-40", "41-60", "61-80", and "80+".

ReMIND

Search chats

New chat

No chats yet

Question 2 of 5

What is age of patient?

0-18

19-40

41-60

61-80

80+

Fig 4.2.10: Security Question 2

ReMIND: Generative AI based Memory Rebuilder

The screenshot shows the ReMIND application interface. On the left is a sidebar with a search bar labeled "Search chats", a "New chat" button, and a message "No chats yet". The main area displays "Question 3 of 5" with a progress indicator. The question is "Are you married, single, divorced or lost partner?". Below the question are four radio button options: "Married", "Single", "Divorced", and "Lost Partner".

Fig 4.2.11: Security Question 3

The screenshot shows the ReMIND application interface. On the left is a sidebar with a search bar labeled "Search chats", a "New chat" button, and a message "No chats yet". The main area displays "Question 4 of 5" with a progress indicator. The question is "Do you have any children? If yes, describe daughters and sons.". Below the question is a text input field with the placeholder "Type your answer here...". At the bottom of the question card is a "Next" button.

Fig 4.2.12: Security Question 4

ReMIND: Generative AI based Memory Rebuilder

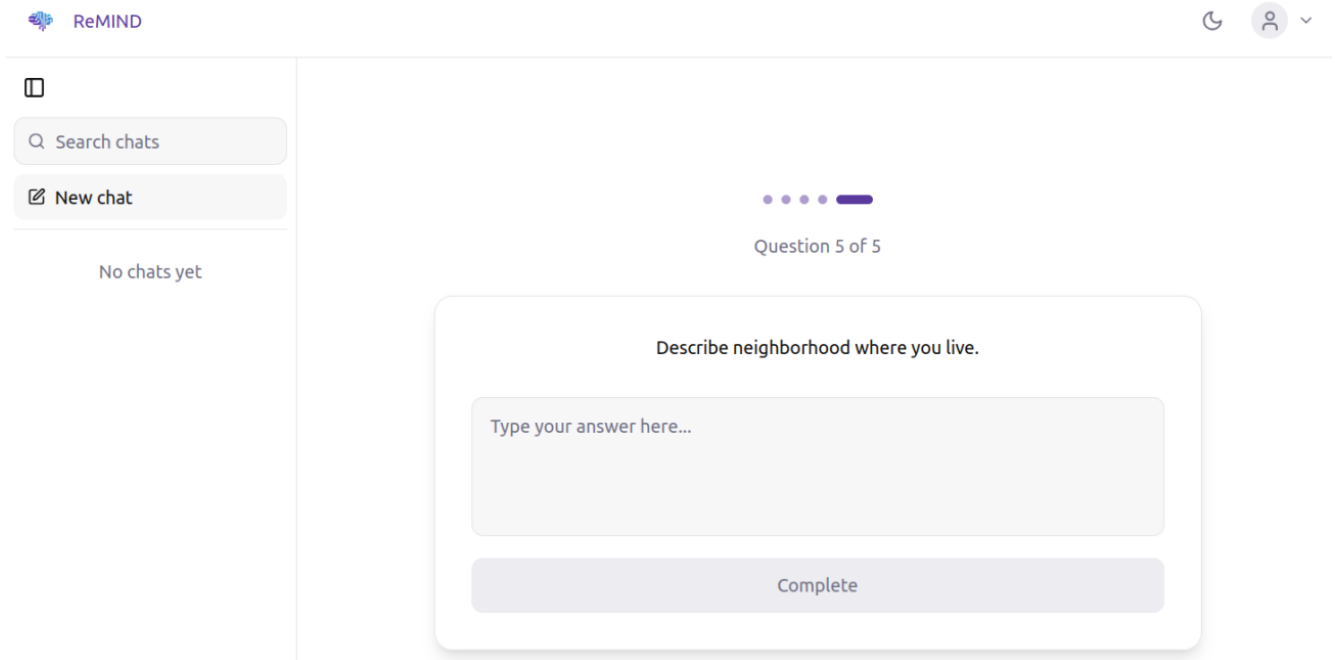


Fig 4.2.13: Security Question 5

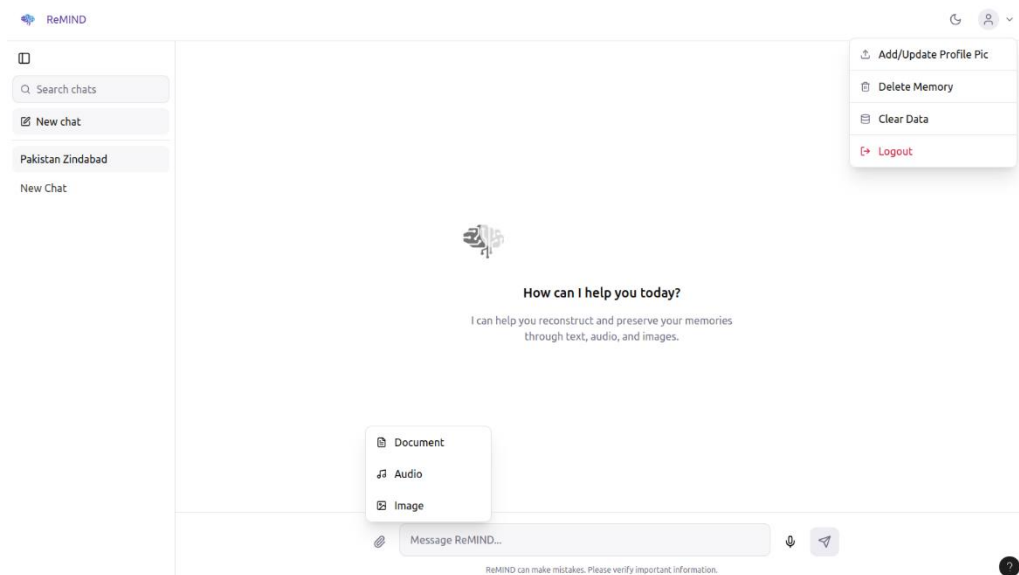


Fig 4.2.14: Chatbot, Main UI in light mode

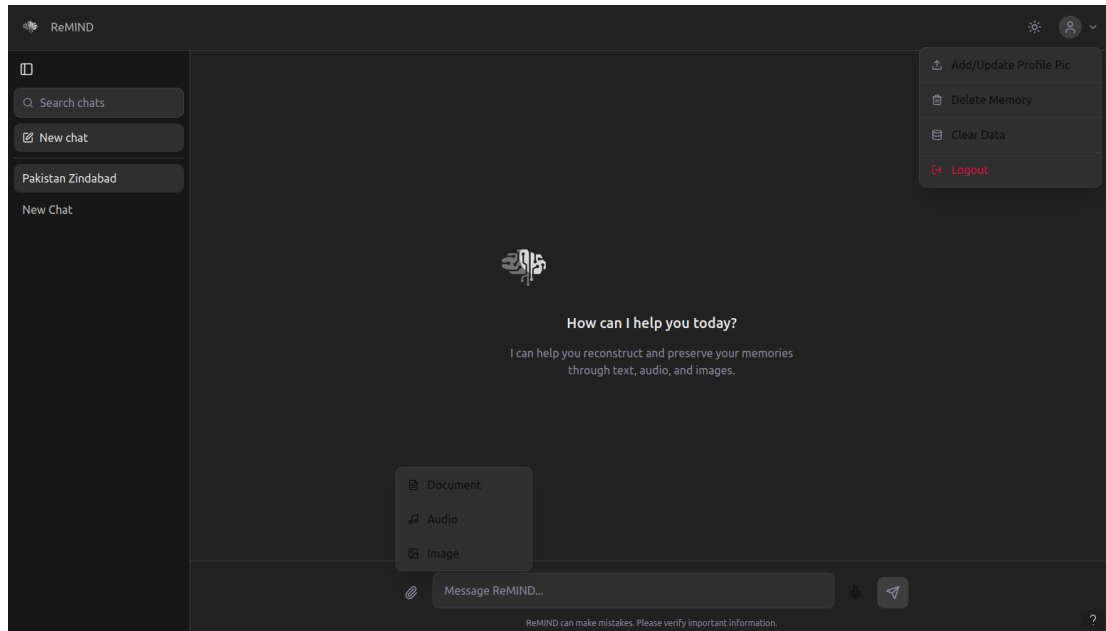


Fig 4.2.15: Chatbot, Main UI in dark mode

4.3 Additional Information

The system design of ReMIND is guided by iterative prototyping using Figma, allowing early validation of user flows, accessibility, and elderly-friendly interaction patterns. Design decisions are continuously refined based on usability feedback, ensuring visual simplicity and clarity. The workflow follows an incremental development approach, enabling modular testing, realistic demonstrations, and smooth integration of AI services throughout development.

5. Revised Project Plan

Currently our project has entered the development phase. Until now we have done research, understood requirements, and project details. We have completed the design process, including sketching, planning, prototyping, official designs, and testing them. Now we are onto front-end and back-end architecting and planning and will shortly start execution. Here is our revised project plan shown using the Gantt chart.

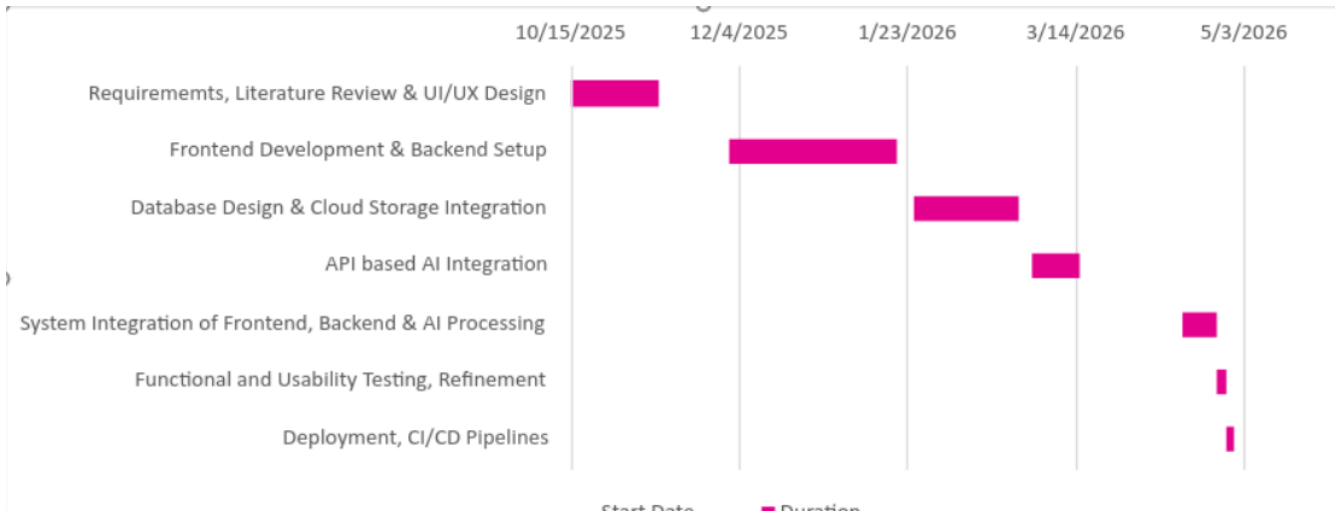


Fig 5.1: Gantt chart

6. References

- [1] Alzheimer’s Disease International, *World Alzheimer Report 2022: Life after Diagnosis — Navigating Treatment, Care and Support*, Alzheimer’s Disease International, 2022. [Online]. Available: <https://www.alzint.org/resource/world-alzheimer-report-2022>
- [2] World Health Organization, “Dementia,” WHO Fact sheet, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/dementia>
- [3] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed., Pearson, 2020.
- [4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
- [5] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. (Draft), 2023. [Online]. Available: <https://web.stanford.edu/~jurafsky/slp3/>
- [6] A. Radford et al., “Whisper: Robust Speech Recognition via Large-Scale Weak Supervision,” OpenAI, 2022. [Online]. Available: <https://openai.com/research/whisper>
- [7] X. Wang, L. Xie, C. Dong, and Y. Shan, “Real-ESRGAN: Training Real-World Blind Super-Resolution with Pure Synthetic Data,” *arXiv preprint arXiv:2107.10833*, 2021. [Online]. Available: <https://arxiv.org/abs/2107.10833>
- [8] OpenAI, “GPT-4 Technical Report,” OpenAI, 2023. [Online]. Available: <https://openai.com/research/gpt-4>
- [9] Microsoft Research, “VALL-E: Neural Codec Language Models for Expressive Text-to-Speech,” Ren et al., 2023. *arXiv preprint*. [Online]. Available: [https://arxiv.org/abs/2304.02341\[10](https://arxiv.org/abs/2304.02341[10)
- [10] ElevenLabs, “Text-to-Speech and Voice AI,” product site, 2024. [Online]. Available: <https://elevenlabs.io>

[11] Descript, “Overdub — Voice Cloning and Audio Editing,” product page, 2023. [Online]. Available: <https://www.descript.com/overdub>

[12] MyHeritage, “Deep Nostalgia — Animate Faces in Old Photos,” MyHeritage blog, 2021. [Online]. Available: <https://www.myheritage.com/deep-nostalgia>

[13] D-ID, “Personal AI avatars for scalable video creation,” product site, 2023. [Online]. Available: <https://www.d-id.com>

[14] A. H. Smith, “Assistive technologies for people with dementia: A review,” *Journal of Assistive Technologies*, vol. 14, no. 2, pp. 85–98, 2020.

[15] PostgreSQL Global Development Group, “PostgreSQL Documentation,” 2024. [Online]. Available: <https://www.postgresql.org/docs/>

[16] Firebase Documentation — Google, “Firebase Storage,” 2024. [Online]. Available: <https://firebase.google.com/docs/storage>

Appendix A: Glossary

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name

Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
...					

Table 1: List of non-trivial defects

This document has been adapted from the following:

1. Previous project templates at UCP
2. High-level Technical Design, Centers for Medicare & Medicaid Services. (www.cms.gov)